

Course Group Assignment: Detecting Robot Bidders in Online Auctions

Group ID: 4

WQF7008 Practical Deep Learning, OCC1

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1. Introduction

Internet auction platforms have become essential tools for online trading, which allow online buyers and sellers to participate in a competitive marketplace. Online auctions, however, can also be affected by unscrupulous bidding behaviour, especially because they are transparent and convenient. In auction fraud research, shill bidding and collusive bidding are often cited as actions that can distort the outcome of an auction and decrease the level of fairness for genuine bidders (Trevathan & Read, 2021; Gerritse & van Wessenbeeck, 2024). The problems are serious, not just with regard to the final price, but also because fake or manipulative bidding can reduce users' trust in the reliability and fairness of the auction platform.

Suspicious bidders are not easy to recognise as fraudulent accounts could try to behave like legitimate bidders. Another way robot or suspicious bidders could get away with it is by concealing their activities within regular bidding behaviour. However, recent attempts to detect auction fraud have shown that there is not always a single, simple rule to follow, and repeated patterns of behaviour and auction activity can also be analysed (Trevathan & Read, 2021; Gerritse & van Wessenbeeck, 2024). Thus, a data-driven method is appropriate, as it enables the model to learn from past bidding patterns that are useful.

To facilitate training of the models used in this project, the raw auction records are transformed into a structured format at the bidder level. This is important because, in the original bids, there are many transaction-level records, but the prediction has to be made at the level of the bidder. The learning problem is made more appropriate for tabular deep learning models by summarising the bidding activity into meaningful behavioural features. Tabular data is also one of the most common formats of data in the real world, but it also has unique properties that make it difficult to process for machine learning and to design models for. Feature interactions can be complicated and non-linear (Borisov et al., 2024; Fan & Waldmann, 2024).

For this project, the main evaluation metric for the model is AUC-ROC, which has been set in the assignment specification. AUC-ROC is suitable because it examines the ability of a model to discriminate between suspicious bidders and normals of decision threshold bidders across different

levels. In addition, indicators such as precision, recall, F1-score, and PR-AUC are viewed as more comprehensive measures of detection performance, especially for cases where the target class is unevenly distributed in the data. Recent imbalanced learning literature also notes that in such scenarios, accuracy might not be sufficient, as it may hide poor detection of the minority class (Chen et al., 2024).

To sum up, this report outlines the whole process of developing the robot bidder detection system, from data preprocessing to feature engineering, and from designing the architecture of the model to training the model, from the evaluation of the model to the analysis of the performance, as well as future directions for improvement and conclusion. The objective is not only to learn how to develop a predictive model but also to learn about the kinds of behavioural and structural signals that can be used to identify automated and/or suspicious bidding activity in online auction data.

2. Problem Statement

Online auction platforms may be affected by fraudulent bidding behaviour, including robot bidding and shill bidding, where fake or suspicious accounts are used to influence the auction process and increase prices unfairly (Trevathan & Read, 2021). In this project, the problem is formulated as a binary classification task, where each bidder is classified as either human or robot based on behavioural bidding data provided in the Facebook Recruiting IV: Human or Robot? competition dataset (*Facebook Recruiting IV: Human or Robot?*, 2015). The main challenge is that suspicious behaviour is usually not obvious from one bid only. Instead, it appears through patterns such as bidding frequency, timing behaviour, device usage, IP usage, and participation across auctions. In addition, fraud detection problems are often imbalanced, where suspicious bidders are fewer than normal bidders, making the classification task more difficult (Gerritse & van Wesenbeeck, 2024). Therefore, this project aims to develop a deep learning model that can effectively distinguish robot bidders from human bidders using bidder-level behavioural features.

3. Model Architecture and Justification

3.1 Architecture

3.1.1 Tabm

TabM is added as a contemporary tabular deep learning model for enhancing performance of multilayer perceptron based models using parameter efficient ensembling. TabM is an approximation for an ensemble of multiple MLP-like predictors as opposed to a single MLP, which makes it more flexible. Yury Gorishniy et al. (2025) call this design "parameter-efficient" to reap the ensembling benefits without training numerous independent neural networks.

The backbone of TabM is an MLP. The feature vector that is sent to the bidder is fed into neural layers that capture non-linear interactions between behavioural variables like bid frequency, diversity of auctions, IP usage, device reusability, timing behaviour and burst bidding indicators. Unlike a single deterministic representation, TabM makes several internal predictions of each input sample. These predictions serve as a kind of ensemble that can help decrease the variance of the model and increase its robustness, while keeping the training and the inference cost low.

The design is crucial due to potential instability and overfitting that can occur with tabular deep learning models, particularly in scenarios with restricted labelled data. TabM achieves these more efficiently than deep ensembles, by sharing most parameters among implicit ensemble members, while maintaining some regularising effect. The model is well suited for structured datasets where the quality of the model relies on capturing complex interactions between features without becoming too complex.

3.1.2 TabR

The TabR architecture consists of three main components: a neural encoder, a retrieval module, and a prediction head. The neural network consists of two parts: a bidder-level feature vector inputted into a neural encoder that maps an input into the latent representation space. This representation enables the model to discover non-linear connections between the behavioral

variables, e.g., the interaction of bidding frequency, timing regularity, device reuse, and the diversity of auctions in which the device is used.

Secondly, TabR fetches training samples which are near the target bidder in the learned representation space. These retrieved neighbours act as further context for the prediction. The main difference is that TabR is not a multilayer perceptron. Unlike a standard neural network, after training TabR enables the prediction to be affected by similar cases in the past. Yury Gorishniy et al. (2024) call the retrieval mechanism "parametric neural learning plus nearest-neighbour information," a technique for enhancing tabular deep learning.

Lastly, the representation of the target bidder and the information retrieved from the neighbours are fused together and fed into a prediction head. The output is a probability value, i.e. the probability of the bidder being in the robot class. Thus TabR is able to learn both global feature interactions and local similarity patterns in the training data.

3.1.3 SAINT

The SAINT model represents a significant advancement in deep learning for tabular data that incorporates both column-wise and row-wise attention mechanisms (Somepalli et al, 2021). This allows SAINT to capture intricate relationships within individual data samples (features) and across different samples within a batch. SAINT directly addresses the limitations of traditional machine learning models and simpler deep learning architectures on structured data.

Traditional gradient-boosted decision trees (GBDTs) have been dominating tabular data tasks because of their efficiency and non-linear interactions modelling. However, when dealing with high-dimensional feature spaces and complex inter-sample dependencies, they fail. Yet most modern datasets are complex and intricate. Deep learning architectures, like Transformer, have shown promise in handling such complexities in other domains like natural language processing and computer vision. According to Somepalli et al. (2021), SAINT extends this paradigm to tabular data by incorporating:

- **FeatureTokenizer** to embed scalar input
- **CLS token** for global representations

- **SAINTLayer blocks** that perform both column attention and row attention.

Recent surveys on digital twins and anomaly detection have highlighted SAINT's superior performance in complex environments compared to traditional neural architectures (Wang et al., 2023).

3.1.4 Spatio-Temporal Hypergraph Neural Network (ST-HGNN)

The Spatio-Temporal Hypergraph Neural Network (ST-HGNN) is designed to capture both the individual characteristics of a bidder and the complex, high-order group dynamics they participate in. Traditional graphs connect two nodes with a single edge, which limits the ability to model multiple users sharing a single resource. To overcome this, ST-HGNN employs a hypergraph structure where a single hyperedge can connect multiple bidders who share the same IP address, device, or participate in the same auction (Liu et al., 2025).

The architecture is built upon a dual-branch framework:

- **Structural (Spatial) Branch:** This branch utilizes stacked Hypergraph Convolution (HypergraphConv) layers to process the interconnected network of bidders. By passing information through these hyperedges, the model captures guilt-by-association signals, effectively identifying communities or rings of suspicious accounts that operate together.
- **Behavioural and Temporal Branch:** In parallel, an MLP processes the static, individual behavioural traits of each bidder. For time-series patterns, a Transformer-based temporal encoder analyses sequence-level features, specifically the bidding burst ratios across different temporal strips to capture how a bidder's activity evolves over time.

Finally, a Squeeze-and-Excitation (SE) gate is used to dynamically fuse these distinct views. The SE gate evaluates the importance of the community context versus the individual's behavioural/temporal traits, intelligently weighting and combining them into a unified representation before passing it to the final classification layer.

3.1.5 Spatio-Temporal Multi-View Hypergraph Neural Network (ST-MV-HGNN)

The Spatio-Temporal Multi-View Hypergraph Neural Network (ST-MV-HGNN) is built upon the Hippocampus-Inspired Multi-View Hypergraph framework (HIMVH) proposed by Cui et al. (2026).

It is specifically engineered to combat “fraud camouflage,” where automated bots alter their individual behavioural signatures to mimic human bidding patterns. To expose these disguised entities, the architecture constructs three separate, parallel hypergraph views representing distinct relational spaces: an Auction View, an IP View, and a Device View. In each view, the specific resource acts as a hyperedge that binds multiple bidders (nodes) together, capturing high-order structural dependencies.

The core predictive power of the ST-MV-HGNN lies in its Cross-View Inconsistency Perception module, which mimics the novelty-detection mechanism of the biological hippocampus. The model processes the three relational views simultaneously through multi-view hypergraph convolutions while running a parallel behavioural branch via a standard Multi-Layer Perceptron (MLP) to capture static bidder attributes. A temporal encoder utilizing a Transformer architecture with dual-pooling tracks the chronological sequence of bidding activities over five distinct time strips. The architecture then uses a “Match-Mismatch” gating logic to explicitly compute structural discrepancies and scene conflicts across the different resource views. If a bidder exhibits perfectly normal individual features but reveals high-order collaborative anomalies within the shared network views, the inconsistency is detected, effectively stripping away the fraud camouflage.

3.2 Justification

3.2.1 TabM

TabM is suitable for this dataset, since the robot bidder detection task was transformed to a bidder-level tabular classification task. The bids are represented by engineered behavioural features derived from raw auction data: Bidding volume, participation in auctions, diversity of devices, diversity of IPs, entropy in the country of the bidders, timing gaps, and indicators related to bursts. These variables don't necessarily work in isolation the more likely way to produce robot-like behaviour is to have a combination of these variables.

In this context, TabM is appropriate for the architecture, as the ability of MLP-based models to learn non-linear interactions between continuous and encoded categorical features is suitable for this context. Meanwhile, its parameter-efficient ensemble structure provides more stable prediction than a single neural network. It is helpful for the Facebook Recruiting IV: Human or Robot dataset since there is a small labelled training set, and it is extremely imbalanced, with only a small percentage of samples representing robot bidders.

TabM also offers a useful comparison against both classical machine learning models and other tabular deep learning models. The tree-based models (Random Forest, LightGBM) are strong baselines for tabular data and the previous research indicated that on some structured datasets, the deep learning did not outperform the aforementioned models (Shwartz-Ziv & Armon, 2022; Grinsztajn et al., 2022; Yury Gorishniy et al., 2024). Hence, TabM can play a role in this study by determining whether a newer model with efficient ensembling, based on MLP, can achieve competitive predictive performance on the behavioural fraud detection data.

3.2.2 TabR

This dataset can be used by TabR, which converts the initial transaction-level bidding records to tabular features at the bidder level. The behaviour variables to describe each bidder are bid frequency, participation in the auction, timing behaviour, device use, IP diversity, country entropy, and burst bidding indicators. These features represent various aspects of the bidder's behaviour, but the robot class cannot be determined based on any single feature alone. Instead, the predictive

signal will more probably be found in between-features combinations and in a bidder similarity with other bidders characterized by similar behavioural profiles.

In this situation the retrieval mechanism in TabR is important, as it enables the model to compare a target bidder with other bidders it has seen during training. The retrieved neighbours can give more context information for the classification, if a bidder has similar behaviour to the labelled robot bidders. This is relevant and important in the context of the Facebook Recruiting IV: Human or Robot dataset, because it is possible that bidders who are suspicious can be similar to normal users on some individual features, but dissimilar on others on a behaviour level.

Additionally, this dataset is very imbalanced, with a small number of robot bidders represented in the labelled data. In these circumstances, the model should be able to learn the minority-class patterns without being "dominated" by the majority human class. The structure of TabR can thus be useful as the retrieval-augmented structure can potentially reinforce the local evidence of the minority class when there is similar robot-like evidence in the training set. But this is only beneficial if the learned representation space is structured in such a way that behaviourally similar robot bidders are grouped together. A weak representation or a small minority class may lead to a retrieval returning a lot of human neighbours and hence not being very useful.

The addition of TabR also enhances the model comparison in this study. In recent years, special models have been introduced for deep learning on tabular data, e.g., TabNet (Arik & Pfister, 2021), SAINT (Somepalli et al., 2021), FT-Transformer (Gorishniy et al., 2021), TabR (Yury Gorishniy et al., 2024). Thus, TabR is a proper deep learning benchmark to test the ability of retrieval-augmented neural models to outperform traditional machine learning models like Random Forest or LightGBM.

3.2.3 SAINT

The justification for SAINT's architecture lies in its ability to explicitly model both feature-level and sample-level interactions which are crucial in understanding complex tabular data. The FeatureTokenizer and positional embeddings allow the model to treat each feature as a distinct entity. Hidden and minute relationships can be captured easily with its dual attention mechanisms

where most models might consider features in isolation. Particularly, the row attention offers a unique advantage by allowing the model to learn from the collective context of a batch. This is highly beneficial in tasks like bot detection where patterns often emerge from comparing multiple instances (Somepalli et al., 2021; Wang et al., 2023).

This architectural design positions SAINT as a strong candidate for tasks requiring heavy understanding of tabular data. For example, bot detection where subtle patterns across features and samples can differentiate real users from automated agents. Its ability to learn complex relationships and interactions makes it a valuable addition to the deep learning considerations for tabular data.

3.2.4 ST-HGNN

STHGNN was selected for this project because it directly addresses the complex, high-order relationships typical of online auction fraud. Traditional graph models often struggle to capture the reality of guilt-by-association, where multiple suspicious accounts might share the same IP address, device, or target the same auction simultaneously. By representing these shared resources as hyperedges, ST-HGNN can model N-to-N relationships, making it highly robust for detecting community-level fraud and interconnected bot networks that might otherwise evade detection.

Furthermore, ST-HGNN architecture is uniquely suited to handle the evolving nature of social bot behavior over time. Its dual-branch design allows it to process not only the structural group correlations via hypergraph convolutions but also the individual behavioural traits via MLP and sequence-level temporal features via a Transformer-based encoder. The inclusion of a SE gate dynamically fuses these perspectives, providing a unified and comprehensive representation that is essential for identifying sophisticated bots in heavily imbalanced datasets.

3.2.5 ST-MV-HGNN

The ST-MV-HGNN architecture is particularly justified for this dataset due to the presence of sophisticated automated accounts that employ “fraud camouflage.” In the Facebook Recruiting dataset, highly deceptive bots often manipulate their individual bidding frequencies and timing sequences to closely mimic legitimate human behaviour. Traditional tabular models or single-view

graphs might easily misclassify these disguised bidders as normal. However, by constructing three distinct relational hypergraphs via Auctions, IPs, and Devices, the ST-MV-HGNN can continuously cross-reference a bidder's structural community context against their isolated behavioural metrics.

The inclusion of the "Match-Mismatch" novelty-aware logic is crucial for exploiting these specific deceptive patterns. If a bidder exhibits perfectly normal individual features but is structurally connected to a dense, suspicious network of shared IPs or devices, the Cross-View Inconsistency Perception module flags this discrepancy as a "scene conflict." This multi-view approach ensures that even the most well-camouflaged bots are detected by validating their individual actions against their broader, high-order network interactions, maximizing predictive accuracy in a heavily imbalanced environment.

3.3 Expected Strengths and Limitations

3.3.1 TabM

The key advantage of TabM is that it is simple (tabular learning with MLP) and strong (ensembling). Because of the multiple predictions that the model has to make, it can lower the reliance on a single learned decision function. This can be useful when identifying robot bidders, as unusual bids can manifest themselves in various ways, including bidding too often, at the wrong time, with the same device or with different IPs.

Another benefit is that its usage is efficient. A typical approach would be to train each of the separate neural networks, but this would be costly in terms of both training cost and inference time. To make the TabM more practical for tabular modelling, most parameters are shared across implicit ensemble members, though retaining the ensemble-like behaviour (Gorishniy et al., 2025).

TabM has drawbacks, though. Does not explicitly model relationships between bidders, auctions, IP address and devices. Therefore, it mainly depends on the quality of the engineered bidder-level features. TabM might not be able to retrieve the important relational signalling directly if it is lost in aggregation. Furthermore, there is an imbalance in the dataset, so performance of the model should be assessed based on minority class detection, not just accuracy. The rare class of robots is of particular importance in this context, as it can be evaluated more clearly using precision-recall based evaluation (Saito & Rehmsmeier, 2015).

3.3.2 TabR

The main advantage of TabR is that it can learn representations and compare instances at the same time. Useful for tabular fraud detection, suspicious activity may manifest itself in local clusters rather than in a single pattern. TabR can recover similar examples based on the neighbourhood information for predicting the example, and potentially learn behavioural patterns that are hard to be captured by a global model.

There are some drawbacks to TabR, though. The effectiveness is highly dependent on the quality of the learned representation space. It is important that the retrieved neighbours be useful and are

not useful if the encoder does not group together robot bidders that are similar in terms of their behavior. Furthermore, retrieval-based models may be sensitive to noisy or weakly separable features, particularly when the number of minorities is limited. Given that a large percentage of the data is labelled by humans, the model might end up picking up mostly human neighbours, unless the space of representation of the data separates suspicious behaviour explicitly. The limitation is crucial because the classification is balanced and should be carefully assessed beyond the notion of accuracy; and the precision-recall based measures are often more informative for minority-class detection (Saito & Rehmsmeier, 2015).

3.3.3 SAINT

SAINT offers distinct advantages in modelling complex tabular data using its attention mechanisms, but it also comes with higher computational cost and data requirements.

SAINT's column attention mechanism effectively models intricate non-linear interactions between features within a single data instance. This is important for identifying subtle, multi-faceted indicators of bot behaviour while surpassing the capabilities of simpler Multi-Layer Perceptrons (MLPs). The unique row attention mechanism also allows SAINT to learn from relationships between different samples in a batch. In bot detection, this means the model can contextualize a bidder's behaviour relative to other bidders, patterns that are more visible when comparing multiple instances. Besides that, the integration of Mixup augmentation during training and DropPath (stochastic depth) regularization enhances the model's robustness to noise and reduces overfitting.

As for the limitations of SAINT, the attention mechanisms introduce a computational complexity of $O(N^2)$ with respect to the batch size (N). This can cause high computational costs and memory requirements when dealing with large datasets or batch sizes. SAINT also requires careful tuning of numerous hyperparameters such as embedding dimensions, number of layers, number of attention heads etc. where choices can significantly impact performance and output results. Fan & Waldmann (2024) argue that deep learning models like SAINT generally require a lot of data to effectively learn complex patterns and generalize well. If a smaller dataset is used, it may underperform simpler models like GBDTs. In terms of interpretability, the complexity of transformer architectures can make direct interpretations of the model's internal decision-making

process challenging. Even though SHAP values can provide insights into feature importance, it is still arguably difficult compared to more transparent models like decision trees.

SAINT is a powerful tool for bot detection where complex and relational patterns captured are important. The balance between model complexity and interpretability is also a consideration because the more advanced the model, the more difficult to understand it directly (Givisis et al., 2025). The only drawback is the high computations cost, need of hyperparameter tuning, and whether data is sufficient for intricate pattern detection.

3.3.4 ST-HGNN

The primary strength of the ST-HGNN model lies in its exceptional capability to detect guilt-by-association fraud. By utilizing a hypergraph structure, it successfully captures high-order, N-to-N relationships among users who share identical IPs, devices, or auctions patterns that standard Graph Neural Networks (GNNs) often miss. Additionally, the model remains robust against extreme class imbalances by integrating normalized hyperedge weights and focal loss. This allows the STHGNN to effectively track and model the evolving, collaborative nature of social bot networks over time, making it superior for community-level fraud detection.

However, this model introduces a high degree of architectural complexity compared to standard MLPs). It requires meticulous, custom hypergraph construction directly from the raw bidding logs, meaning its overall predictive performance relies heavily on the quality and relevance of these pre-defined hyperedge structures. Furthermore, hypergraph convolutions are notably computationally expensive and memory-intensive, which can pose significant scalability challenges during training when processing massive, high-dimensional auction datasets.

3.3.5 ST-MV-HGNN

The defining strength of the ST-MV-HGNN model is its unparalleled ability to detect “fraud camouflage” in highly imbalanced datasets. By continuously comparing structural community contexts across three distinct resource views (Auctions, IPs, and Devices) against a bidder’s individual behavioural signals, the model excels at exposing discrepancies. When an automated bot successfully mimics human behaviour on an individual level but remains structurally tied to a known fraud ring, the model’s novelty-aware logic flags this “scene conflict.” This multi-view

cross-referencing grants the ST-MV-HGNN exceptionally high precision in identifying sophisticated, evasive bots that would easily bypass single-view graphs or standard tabular models.

However, these advanced capabilities come with substantial computational and architectural trade-offs. The ST-MV-HGNN is extremely computationally intensive, as it must construct, maintain, and process three parallel hypergraph convolutions along with complex gating logic for every single data point. Furthermore, the model's temporal effectiveness is heavily reliant on the specific extraction of sequence-level "burst" features across five high-resolution time strips. This strict dependency on granular timestamp data makes the architecture less flexible and harder to generalize to other fraud detection datasets that may lack detailed, high-resolution temporal logs.

3.4 Summary

Models	Architecture	Justification On Dataset	Strength	Weakness
TabM (Gorishniy et al., 2025)	MLP-based model that makes several parallel predictions with shared weights, then combines them like an efficient ensemble.	Suitable for tabular bidder features with non-linear behavioural interactions.	Strong performance, efficient, and simple official implementation	Still needs tuning and is less interpretable than TabNet (Arik & Pfister, 2021)
TabR (Gorishniy et al., 2024)	A retrieval-augmented tabular deep learning model. It is essentially a feed-forward neural network (MLP-like backbone) with a custom k-nearest-neighbours-like retrieval component in the middle. The retrieval part uses a single-head attention-like module to retrieve useful training samples and improve prediction.	Fits bidder-level tabular features where suspicious behaviour may appear through similarity patterns.	High predictive performance on tabular tasks. The paper reports that TabR achieves the best average performance among tabular deep learning models, reaches state-of-the-art on several datasets, and is more efficient than earlier retrieval-based tabular DL models. It also supports binary classification, multiclass classification, and regression, so it is flexible for many tabular prediction problems.	More expensive than simpler models in practice. Although TabR is stronger than earlier retrieval-based DL models, the paper also states that tree-based models remain a cheaper solution, and TabR is still slower than simple parametric models and GBDT. In addition, reproduction is easiest only under a fixed hardware/software environment, which makes deployment and exact reproduction less convenient than simpler baselines.
SAINT	Extended Transformer encoder with dual-attention mechanism.	Re-derives a focused, semantically meaningful 30-	SAINT can leverage relationships between similar	Intersample Attention requires a batch of samples to calculate

(Fan & Waldmann, 2024)	SAINT projects every feature into a n-dimensional embedding space. Column Attention (Self-Attention) applied across sequences of feature tokens within a single row, allowing the model to learn complex, non-linear correlations between different variables of a single record. Row Attention (Intersample Attention) applies multi-head attention (MHA) across the same feature column across different samples within a batch, allowing the data point to look at other samples to refine its own representation.	feature representation directly from raw bid stream. Keeping the embedding space interpretable and avoids feature leakage from tree-centric engineering choices.	samples, thanks to Intersample Attention, making it highly effective for datasets where the neighbourhood of a point in latent space provides strong predictive signals (Fan & Waldmann, 2024). Due to the self-attention mechanism, it also captures higher-order feature interactions better than standard feed-forward networks (MLPs).	relationships, making predictions for a single new record problematic, if either to disable row attention at inference or provide a reference batch which can introduce inconsistencies. According to Fan & Waldmann (2024), SAINT occasionally outperformed in accuracy and efficiency by LassoNet in specific multi-task genome-wide prediction scenarios.
ST-HGNN	BotSTHCL (Liu et al., 2025): A Community-Aware Spatio-Temporal Hypergraph Contrastive Learning framework. It utilizes a dual-branch design: a structural branch with stacked HypergraphConv layers to capture group correlations and a behavioural branch (MLP) for individual traits. A Transformer-based temporal encoder processes sequence-level bidding features, while a Squeeze-and-Excitation	Effectively captures “guilt-by-association” by representing shared IPs, devices, and auctions as hyperedges to model high-order N-to-N relationships, identifying interconnected bot networks that standard graph models miss.	This model is highly effective at identifying “guilt-by-association” by capturing high-order N-to-N relationships between users sharing IPs, devices, or auctions. By using hyperedge weights and focal loss, it remains robust against class imbalance and successfully models the evolving nature of social bot behaviour over time, making it superior to standard GNNs for	The architecture has a high degree of complexity compared to standard MLPs, requiring custom hypergraph construction from raw bidding logs. Furthermore, hypergraph convolutions are computationally expensive on large datasets, and the model’s performance relies heavily on the quality of the pre-defined hyperedge structure.

	(SE) gate dynamically fuses community and individual perspectives into a unified representation.		community-level fraud detection.	
ST-MV-HGNN	HIMVH (Cui et al., 2026): A Hippocampus-Inspired Multi-View Hypergraph model. It features a Cross-View Inconsistency Perception module that analyses separate hypergraph views (Auctions, IPs, and Devices) in parallel. The core mechanism uses “Match-Mismatch” novelty-aware logic to compare structural community context against individual behavioural signals, and includes a Transformer encoder with dual-pooling for temporal sequence modelling.	Ideal for detecting “fraud camouflage” in the highly imbalanced dataset. By comparing structural community context against individual behavioural signals across multiple views (Auctions, IPs, Devices), it uncovers discrepancies where bots mimic human behaviour.	The primary strength lies in its ability to detect “fraud camouflage,” where individual bots mimic human behaviour but remain structurally linked to suspicious networks. By identifying “Scene Conflicts” (discrepancies between different resource views), it achieves higher precision in detecting sophisticated bots that attempt to hide within legitimate user groups.	This model is extremely computationally intensive, as it must process three parallel hypergraph views and complex gating logic for every data point. Additionally, its effectiveness depends on the specific extraction of temporal “burst” features across five time strips, making it less generalizable to datasets lacking high-resolution time logs.

4. Data Preprocessing Methods

The successful application of machine learning models on Facebook Recruiting IV dataset is determined by the effective data preprocessing. This is because raw bid-level event logs and bidder-level information are transformed into a structured, clean, and semantically rich feature set. The preprocessing pipeline is designed to capture the diverse behavioural patterns of both human and automated agents.



Figure 4.1: This dashboard provides a high-level overview of the raw dataset comparing Facebook human and bot bidding activity, including visualizations for class balance, bids per bidder, top countries, top devices, merchandise distribution, and bid timestamps.

Based on Ibrahimi et al. (2023) and Mu et al. (2026), they mentioned that effective preprocessing prevents models from “falling prey to garbage-in, garbage out” scenarios in tabular data preparation. The general methods applied across the modeling framework include:

4.1 Data Cleaning and Imputation

Duplicate Removal

Exact duplicate rows based on unique identifiers are removed from the bid-level event logs to prevent double-counting of bids, which could otherwise bias the engineered features.

Missing Value Imputation

To preserve all data rows, categorical null values in columns like country, URL, and IP are filled with “UNKNOWN” values. It allows the model to treat missing values as an informative category. For numerical features, median imputation is used to prevent target leakage by calculating median solely from training distribution.

Duplicate Rows	No duplicate rows within train.csv, test.csv, and bids.csv
Missing Values	8,859 missing values in bids.csv for country (0.1157%)

4.2 Feature Engineering

Volume and Diversity

Metrics such as total bid counts and the number of unique auctions, devices, countries, IP addresses, and referral URLs participated in by each bidder.

Behavioural Rates

Features such as bids per auction, IPs per bid, and countries per bid are normalized to expose the density of activity relative to its scale. The scale is often highly discriminative for bot detection.

Timing and Temporal Dynamics

Features that capture the time span of activity, mean and standard deviation of bid timestamps, and inter-bid gap statistics. A high proportion of bids with identical timestamps is a strong indicator of bot trading activity.

Entropy and Diversity

Shannon entropy is computed for categorical dimensions like country and device category. High entropy signals diverse human-like variation, while low may indicate mechanical repetition.

Burst Activity

Maximum and mean bid counts within varying temporal windows are extracted to capture rapid, short-term spikes in bidding activity.



Figure 4.2.1: A grid of histograms displaying the feature distributions for both human and bot classes across six variables, such as log bid count, log unique IPs, and zero-interval ratio.

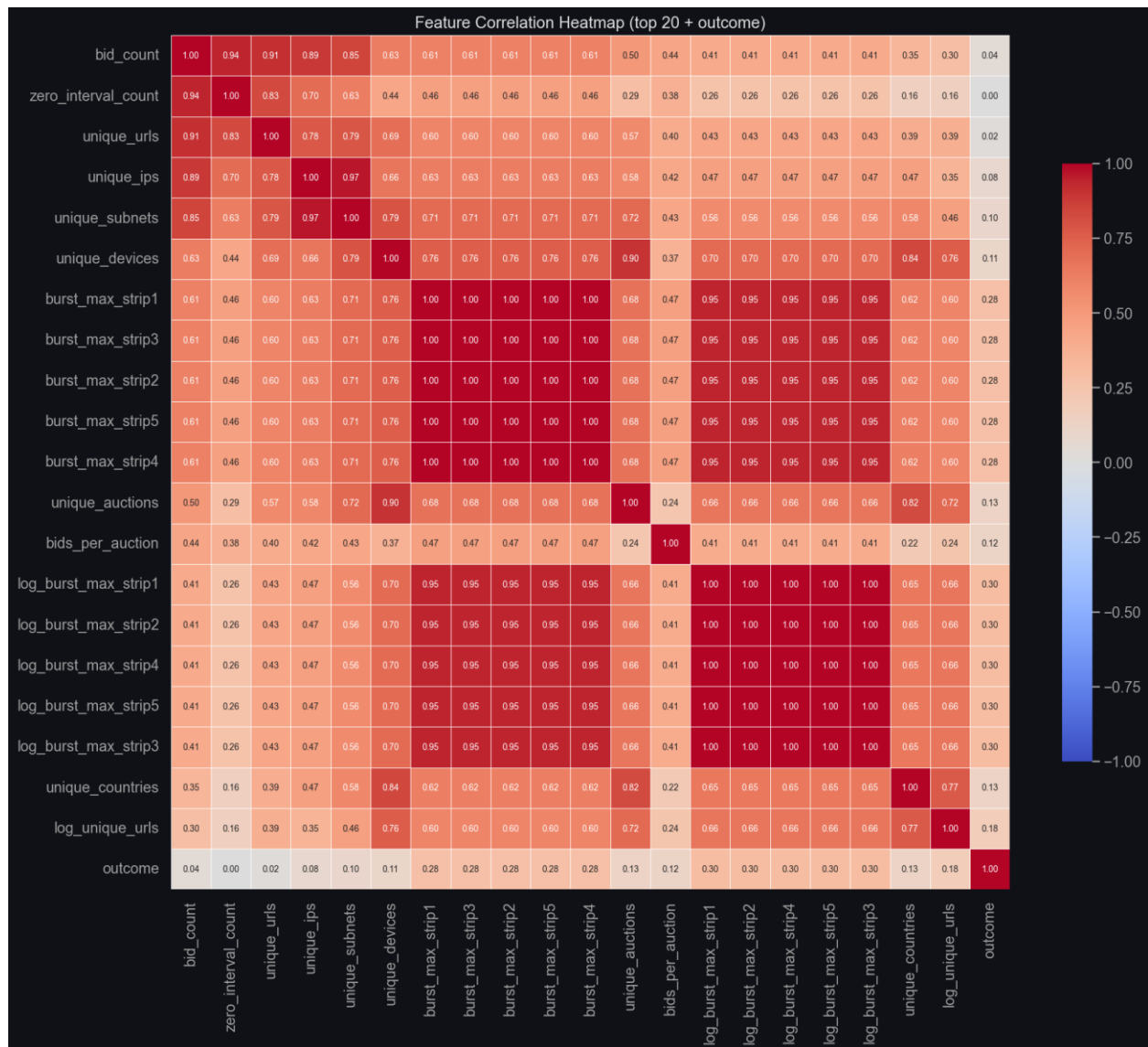


Figure 4.2.2: A correlation heatmap detailing the relationships between the top twenty predictive features such as bid count and unique IPs alongside the final target outcome.

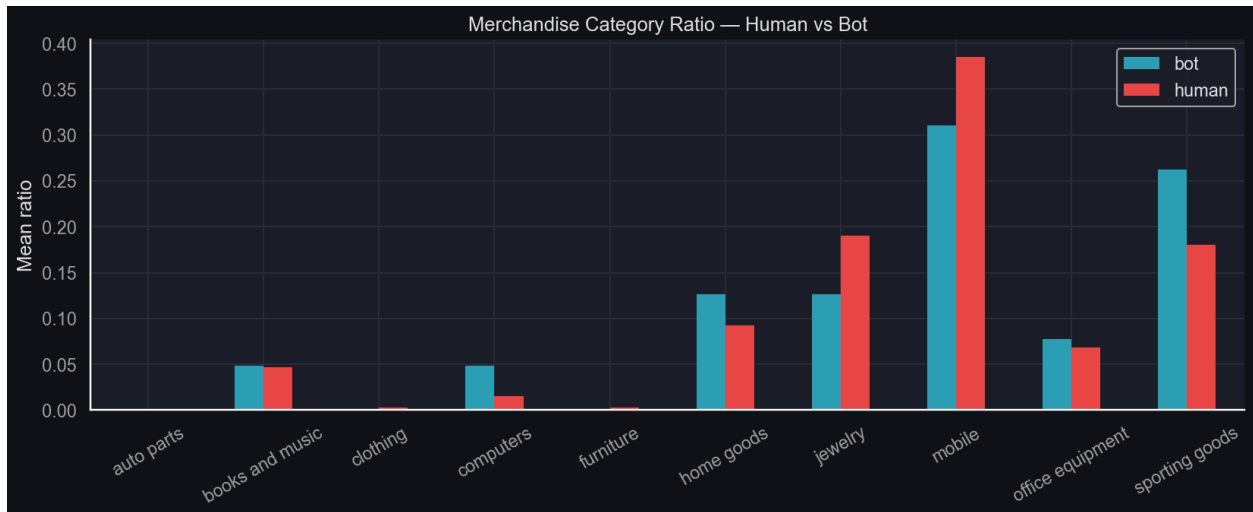


Figure 4.2.3: A grouped bar chart comparing the mean ratio of bids placed across various merchandise categories such as mobile, jewelry, and sporting goods by bots versus humans.

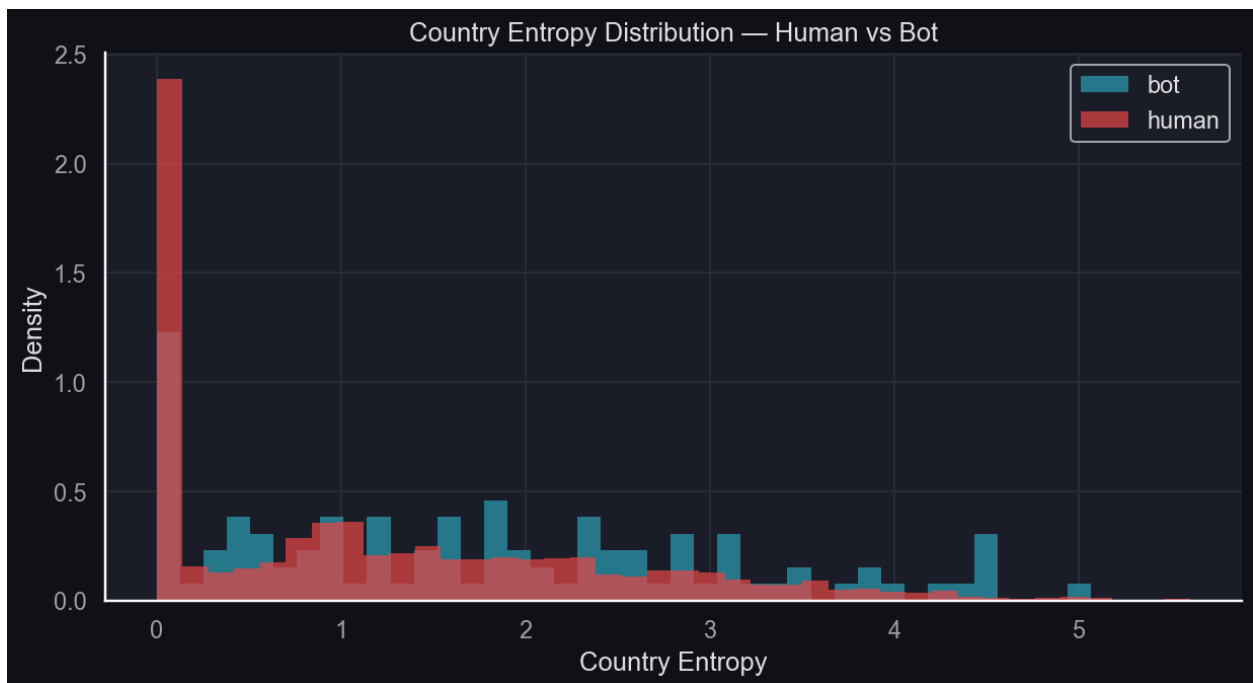


Figure 4.2.4: A density histogram showcasing the distribution of country entropy for humans and bots, highlighting a distinct, sharp concentration near zero for human users compared to a wider spread for bots.

4.3 Data Transformation and Scaling

Log-Transforms

Log-plus-one transforms are applied to right-skewed volume and time features to improve numerical stability and make visual comparisons more balanced.

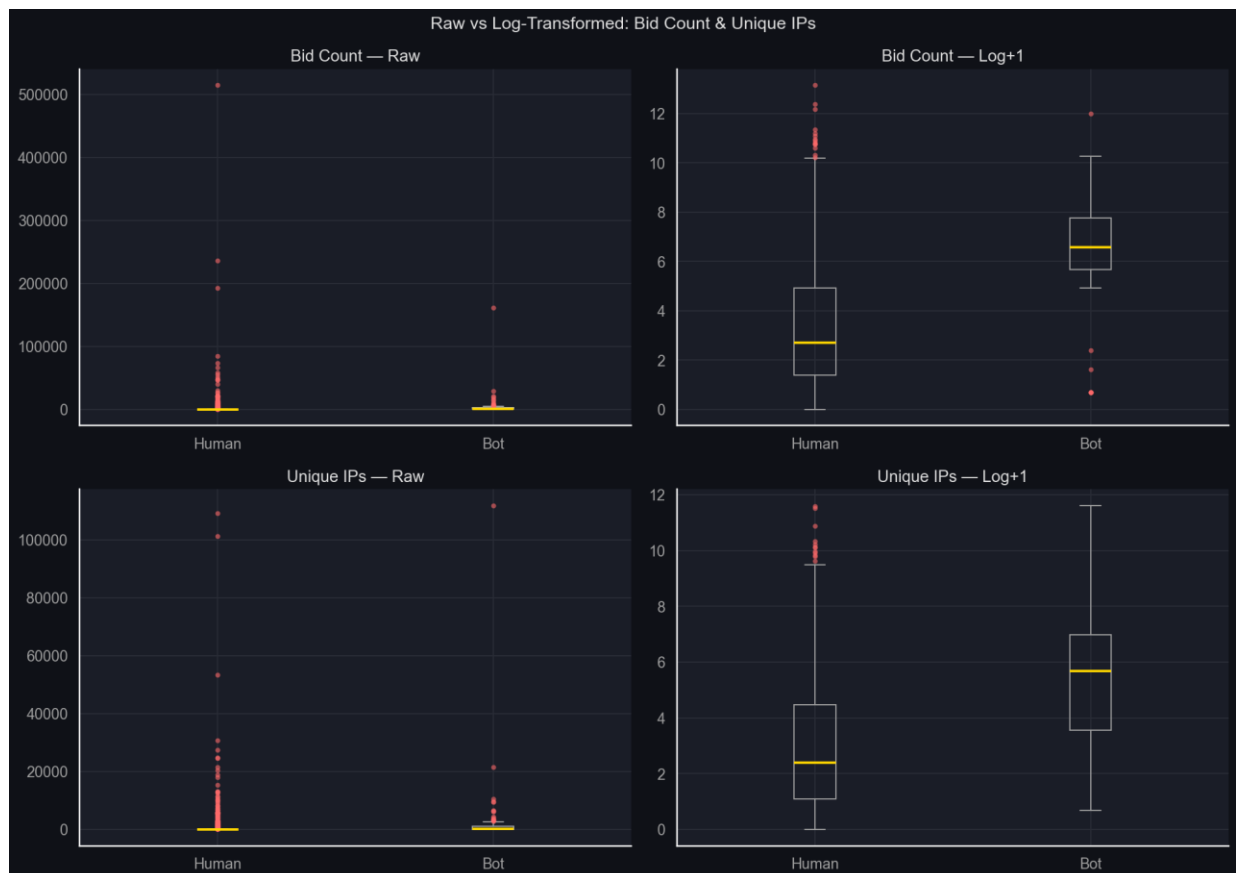


Figure 4.3.1: Boxplots illustrating the effect of log transformations on highly skewed features like bid count and unique IPs, grouped by human and bot classifications to show how outliers are mitigated.

Standardization

Numerical features are scaled to achieve zero mean and unit variance using `StandardScaler`. This is crucial for neural network models that are sensitive to the absolute scale of input features. The general preprocessing methodology stresses on the transformation of raw, high cardinality event logs into a fixed-length feature vector for each bidder. The pipeline can capture the fundamental

differences between human and bot behaviour by focusing on volume, diversity, timing and entropy. The use of sentinel values for missing data and median imputation from the training set ensures a robust and leakage-free preparation process (Ibrahimi et al., 2023; Mu et al., 2026). These methods provide a comprehensive behavioural profile of each bidder to serve as a solid foundation for any downstream classification model.

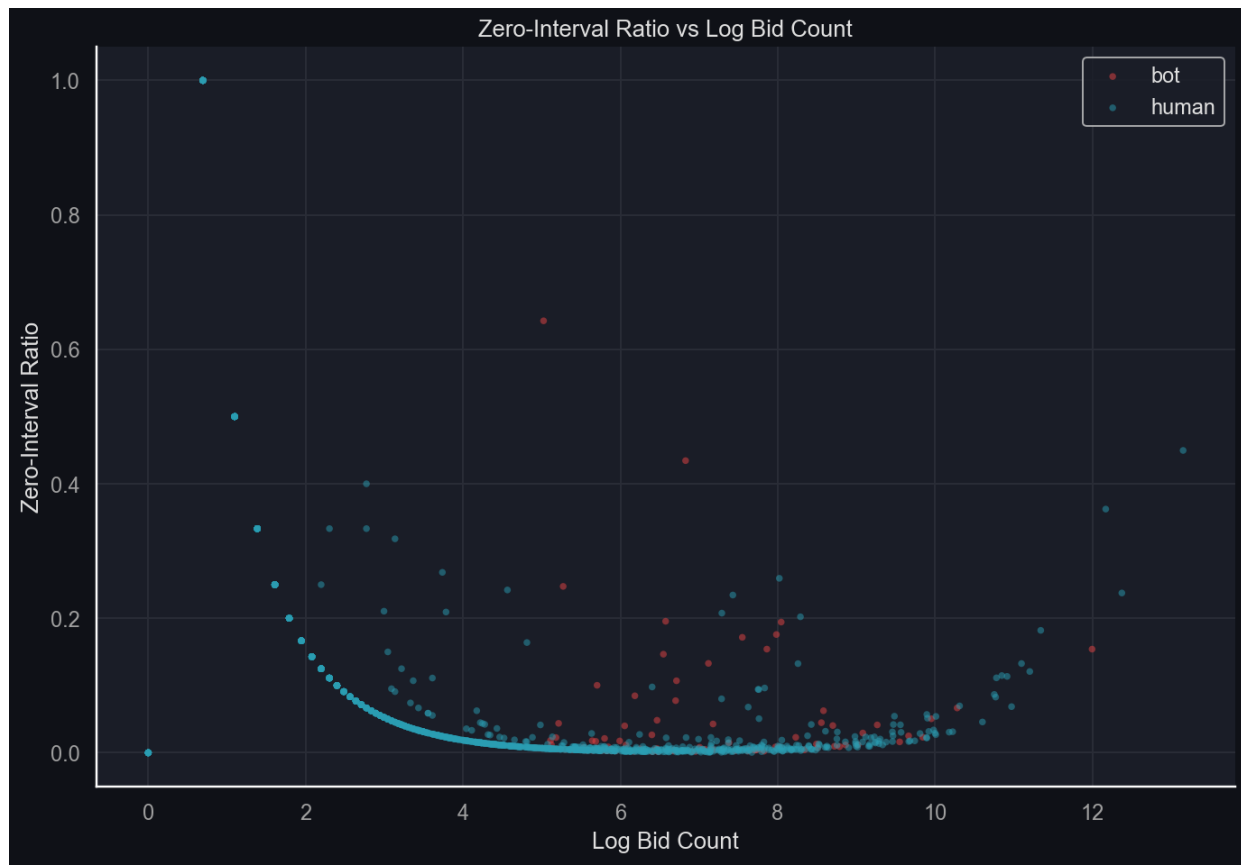


Figure 4.3.2: A scatter plot of the zero-interval ratio against the log bid count, demonstrating how bot and human bidding behaviours separate and cluster across these two dimensions.

5. Model Training and Evaluation

5.1 Training Setup

5.1.1 TabM

The TabM model is trained with a binary classification objective for the classification of human bidders from robot bidders. Class imbalance is solved by positive-class weighting because the robot class accounts for only about 5.12% of the labelled dataset. This weighting is an increase to the influence of robot bidders without the need for oversampling when optimising. This is necessary, because oversampling can result in redundant minority class patterns and class weighting can maintain the original training set distribution.

Optimizer and Scheduler

AdamW optimizer is used as it uses decoupled weight decay that achieves better regularisation than regular L2. This is a good choice for a neural tabular model where the number of labelled bidders is relatively small and overfitting can be easily achieved. A ReduceLROnPlateau scheduler is used to decrease the learning rate if the validation loss doesn't change anymore. This enables the model learning at a high rate, and then slowly slow that rate during the optimization process.

Class Imbalance Handling

During loss optimisation, the weighting used for robot bidders is 0.75 (Positive class). This is needed because otherwise the model would be biased towards the majority-human class. The weighting strategy enables TabM to learn patterns of minority classes whereas maintaining the original distribution of classes in the data set.

Cross-Validation

TabM is validated via stratified 5-fold cross validation to get reliable validation estimates. Stratification makes sure that the minority robot class is present in all folds. This is crucial since a clean random split could lead to some validation folds that do not have adequate bidders from robot clients to ensure a stable performance estimate. The out-of-fold design also ensures that all labelled bidders are given a model that has not been trained on them.

Training Parameters

Setting	Value	Rationale
Folds	5	Standard practice for robust validation on small datasets
Class weighting	Positive-class scale = 0.75	Reduces majority-class dominance during optimisation
Optimizer	AdamW	Provides decoupled weight decay for better regularisation
Learning rate	0.001	Selected after testing 0.001 and 0.0007
Scheduler	ReduceLROnPlateau	Reduces learning rate when validation loss plateaus
Preprocessing	Quantile transformation	Reduces the effect of skewed behavioural features

The best model checkpoint is determined by the validation performance during training, not at the end of the epoch. This ensures that the selected model does not overfit the training fold, particularly if the robot class is small and the performance in the validation fold varies around the fold.

5.1.2 TabR

The TabR model is trained with Binary Cross-Entropy with Logits Loss which is appropriate for binary classification tasks. The model makes a classification of the person from human or robot class. The data set is extremely skewed, and therefore the performance is not only measured in terms of accuracy, but also in terms of threshold independent and minority class sensitive capacity-area-under-the-curve-roc and average precision scores.

Optimizer and Scheduler

Both adaptive learning and decoupled weight decay are implemented in the AdamW optimizer, which is utilized for training TabR. The advantages are suitable for retrieval-augmented neural models, where regularisation is crucial to avoid the latent representation space overfitting the relatively small labelled training set. For this example, a learning rate of 0.001 and a weight decay

of $1e-5$ are used, which enables the model to learn meaningful representations of the bidders, while avoiding the growth of too many impressive parameters.

Retrieval-Based Training

The difference between TabR and traditional neural networks is that TabR fetches similar training examples when predicting. Bidders are first mapped into a latent representation space and then the model uses the contextual evidence of neighboring bidders in the training set. This is helpful for detecting suspicious behaviour in a robot bidder, as it can be compared to other similar historical bidders for easier detection, opposed to only using global interactions between features.

Train-Validation Split

For TabR, an 80:20 stratified train-validation split is used. In both subsets, stratification keeps the ratio of human to robot bidders the same as in the original dataset. This is significant since the class of minority robots is limited and, if the composition of the validation set is unstable, it can result in incorrect performance results. StandardScaler is used to standardise numerical features and OrdinalEncoder is used to encode categorical features. They are fitted to the training split and then applied to the validation and test data.

Training Parameters

Setting	Value	Rationale
Train-validation split	80:20 stratified	Preserves robot-class ratio in validation
Batch size	256	Balances training efficiency and gradient stability
Max epochs	200	Allows sufficient training time for convergence
Early stopping patience	30 epochs	Prevents overfitting after validation stagnation
Optimizer	AdamW	Provides adaptive optimisation with decoupled weight decay
Monitor metric	Validation AUC-ROC	Measures threshold-independent ranking quality

In the training stage, the best TabR checkpoint is recovered based on the validation AUC-ROC. This is essential as long as the training loss continues to reduce even though the validation loss doesn't. The checkpoint with the best performance was taken at the 63rd epoch in the TabR notebook.

5.1.3 SAINT

The SAINT model utilizes Binary Cross-Entropy with Logits Loss, which is suitable for binary classification tasks. Importantly, this loss function combines a per-fold positive class weighting mechanism to dynamically upweight the minority bot class within each cross-validation fold. According to Somepalli et al. (2021), this effectively compensates for the inherent class imbalance without using oversampling techniques.

Optimizer and Scheduler

AdamW optimizer is employed for its decoupled weight decay, which is preferable for Transformer architectures compared to L2 regularization. OneCycleLR learning rate scheduler is used to ramp the learning rate up to a peak and gradually decay it using cosine annealing. This approach acts as an implicit regularizer and leads to faster convergence.

Mixup Augmentation

Per-batch Mixup augmentation is applied to both inputs and soft-targets before each forward pass during training. This technique generates synthetic training examples by linearly interpolating two random samples and their labels. Thus, promoting vicinal risk minimization and reducing overconfidence.

Cross-Validation

To obtain reliable performance estimates, the original class ratio of bots to humans needs to be preserved within each fold, thus a stratified 5-fold cross-validation strategy is implemented.

Training Parameters

During training, the best model state is checkpointed and restored after each fold completes. It is worth noting that row attention is disabled during training when Mixup augmentation is applied. This prevents the corruption of inter-sample patterns by artificial samples.

Setting	Value	Rationale
Folds	5	Standard practice for robust evaluation
Batch Size	512	Chosen to balance computational efficiency and gradient stability
Max Epochs	200	Enough epochs to allow for full convergence
Early Stopping Patience	20 epochs	Prevents overfitting and saves computational resources.
Monitor Metric	Validation AUC-ROC	A threshold-independent metric that directly measures ranking quality

5.1.4 ST-HGNN

The ST-HGNN model is trained with a binary classification objective to distinguish between human and robot bidders. Given the extreme class imbalance in the dataset, standard loss functions often struggle as they become dominated by the majority human class. To address this, the training setup replaces standard cross-entropy with a Focal Loss function, which dynamically scales the loss based on prediction confidence, forcing the model to focus on hard-to-classify robot examples.

Optimizer and Scheduler

The model utilizes the Adam optimizer with a learning rate of 0.005 and a weight decay of $1e-4$ to provide stable parameter updates and prevent the complex hypergraph embeddings from overfitting. A CosineAnnealingLR scheduler is implemented to gracefully decay the learning rate, helping the model settle into optimal local minima during the later stages of training.

Class Imbalance Handling

In addition to the Focal Loss (configured with $\alpha=0.85$ and $\gamma=2.5$), the model handles the sparse and imbalanced nature of the auction graph by applying a logarithmic normalization to the hyperedge weights. This prevents highly active nodes (e.g., massive shared IP networks) from dominating the message-passing updates, ensuring that subtle bot behaviors are not drowned out by sheer volume. Hyperedge dropout (setting 10% dropout to 0.8) is also applied during training to improve generalization.

Cross-Validation

To ensure reliable and robust performance estimates, a stratified 5-fold cross-validation strategy is used. The stratification guarantees that the rare robot class is proportionally represented across all training and validation folds.

Training Parameters

Setting	Value	Rationale
Folds	5	Standard practice for robust validation while preserving class ratios.
Loss Function	Focal Loss ($\alpha=0.85$, $\gamma=2.5$)	Directly addresses extreme class imbalance by down-weighting well-classified majority examples.
Optimizer	Adam ($\text{lr}=0.005$, weight decay= $1e-4$)	Provides efficient gradient descent with light L2 regularization.
Scheduler	CosineAnnealingLR	Smoothly decays the learning rate for better convergence.
Max Epochs	200	Allows sufficient iterations for the hypergraph convolutions to converge.
Early Stopping Patience	25 epochs	Halts training when validation AUC plateaus, preventing overfitting on the training fold.
Monitor Metric	Validation AUC-ROC	Ensures the model checkpoints are saved based on ranking quality rather than threshold-dependent accuracy.

During training, the model evaluates its performance using validation AUC-ROC, and the best model state is checkpointed. If the validation performance fails to improve for 25 consecutive epochs, early stopping is triggered to save computational resources and avoid overfitting.

5.1.5 ST-MV-HGNN

The ST-MV-HGNN model employs a binary classification objective to differentiate between human and robot bidders. To effectively counter the severe class imbalance present in the dataset, the standard cross-entropy loss is replaced with a Focal Loss function (configured with $\alpha=0.85$ and $\gamma=2.5$). This ensures the model heavily penalizes confident but incorrect predictions, forcing the network to concentrate its learning capacity on the hard-to-classify disguised bots.

Optimizer and Scheduler

The multi-view architecture is trained using the Adam optimizer with a learning rate of 0.005 and a weight decay of $1e-4$. To aid the convergence of the complex, multi-branch network, a CosineAnnealingLR scheduler is implemented. This gradually decays the learning rate following a cosine curve, allowing the model to make large initial updates before smoothly settling into optimal local minima.

Class Imbalance and Regularization

In addition to the Focal Loss, the training regimen heavily utilizes hyperedge dropout as a regularization technique. During training, 20% of the hyperedges across the Auction, IP, and Device views are randomly dropped during each forward pass. This prevents the model from memorizing specific, static fraud rings, forcing it to learn generalized structural anomalies and cross-view discrepancies. Furthermore, the hyperedge weights are normalized across all views to prevent massive nodes (e.g., heavily trafficked public IPs) from drowning out nuanced behavioural signals.

Cross-Validation

The model uses a stratified 5-fold cross-validation setup to provide rigorous and unbiased out-of-fold performance estimates, ensuring that the minority robot class is equally represented across all validation splits.

Training Parameters

The training loop continuously monitors the validation AUC-ROC, saving the best-performing model state for each fold. If the validation performance fails to improve for 40 consecutive epochs, early stopping is triggered to prevent the model from overfitting on the training data.

Setting	Value	Rationale
Folds	5	Ensures robust validation while keeping class ratios consistent.
Loss Function	Focal Loss ($\alpha=0.85$, $\gamma=2.5$)	Focuses the learning signal on difficult minority-class examples.
Optimizer	Adam ($\text{lr}=0.005$, weight decay= $1\text{e-}4$)	Provides efficient optimization with decoupled weight decay to prevent overfitting.
Scheduler	CosineAnnealingLR	Smoothly decays the learning rate to ensure stable convergence.
Max Epochs	200	Provides adequate iterations for the multi-view gating logic to optimize.
Early Stopping Patience	40 epochs	A longer patience window allows the complex Cross-View Inconsistency logic enough time to stabilize before halting.
Monitor Metric	Validation AUC-ROC	Used to checkpoint the best model state based strictly on ranking quality.

5.2 Model Configuration

5.2.1 TabM

The TabM configuration describes a neural tabular model of an MLP backbone with parameter-efficient ensembling. The idea is to share most of the parameters of TabM and to generate multiple predictions from the internal predictions. Gorishniy et al. (2025) seek to maintain the power of ensembling while also cutting computational expenses. The following is a summary of the configuration parameters:

Hyperparameter	Value	Rationale
Ensemble members	8	Provides ensemble-like prediction without training separate models
Neural blocks	2	Moderate depth reduces overfitting risk on small data
Hidden dimension	128	Balances model capacity with dataset size
Dropout	0.20	Regularises the MLP and reduces overfitting
Learning rate	0.001	Selected for stable convergence and validation performance
Weight decay	0.001	Controls model complexity through regularisation
Preprocessing	Quantile transformation	Handles skewed numerical feature distributions

The following parameters make TabM a lean but compact model for bid-level tabular data. Parameter-efficient ensembling helps stabilize training from that of a single MLP, while dropout and weight decay decrease overfitting, and validation-based checkpointing prevents overfitting. This setup is appropriate for the data set since there is a small number of labelled bidders and the positive robot class is less abundant.

5.2.2 TabR

TabR is a retrieval-augmented neural model for tabular classification defined by the TabR configuration. The model learns a hidden representation for each bidder and uses the same hidden representation as context for prediction (training examples are used for context). This is done by combining both neural representation learning and nearest-neighbour evidence, which enables the model to leverage global feature interactions and local similarity patterns, says Gorishniy et al. (2024). The configuration parameters that are important are summarized below:

Hyperparameter	Value	Rationale
Latent dimension	192	Defines the representation space for bidder similarity
Context size	96	Number of retrieved neighbours used during prediction
Prediction blocks	1	Limits model complexity and reduces overfitting risk
Context dropout	0.10	Regularises retrieved-neighbour dependence
Learning rate	0.001	Supports stable optimisation with AdamW
Weight decay	1e-5	Provides light regularisation for neural weights
Batch size	256	Balances memory use and training stability
Max epochs	200	Allows the model sufficient time to converge

The parameters enable TabR to learn the representations of the bidders, and access the representations of the neighbouring training samples as supplemental prediction context. The context size of 96 is large enough to give local evidence for retrieval-based prediction, while the context dropout ensures that the model is not too sensitive to a specific set of neighbours. This is

relevant for the recognition of suspicious bidders since they can be similar to human bidders in some features and not in others, when using a wider behavioural similarity.

5.2.3 SAINT

The model configuration defines specific hyperparameters carefully chosen to suit the size and complexity of the dataset. The key configuration parameters are summarized below:

Hyperparameter	Value	Rationale
dim	128	Embedding dimension for each feature. Balances capacity with the ~2,000 training samples.
depth	4	Number of SAINTLayer blocks. Moderate depth avoids overfitting on small datasets.
n_heads	8	Number of attention heads. Allows the model to attend to different feature subspaces simultaneously.
ffn_mult	4	Expansion factor for the hidden layer within the feed-forward network.
attn_dropout	0.1	Applied within the attention modules to prevent overfitting.
mlp_dropout	0.1	Applied within the classification head's MLP for regularization.

These parameters collectively define the SAINT model with approximately 1.3 million trainable parameters. This is considered large enough to model complex interactions within the tabular data without severe overfitting on available training bidders. The selection of dimension, depth, and heads balance the model capacity and the risk of overfitting. The use of dropout in both attention and MLP layers further reinforces the regularization strategy.

5.2.4 ST-HGNN

The ST-HGNN configuration establishes a sophisticated dual-branch architecture designed to process both the structural hypergraph data and the individual tabular features. The goal of this configuration is to balance the computational efficiency required for graph-based learning with the

model capacity needed to detect subtle fraud patterns. The key configuration parameters for ST-HGNN are summarized below:

Hyperparameter	Value	Rationale
Hidden Channels (Dimension)	64	Provides sufficient representational capacity for both the spatial and behavioural branches while keeping the computationally expensive hypergraph convolutions efficient.
HypergraphConv Layers	2	Captures 2-hop community structures (e.g., bidders sharing a device with bidders who share an IP) without causing the over-smoothing common in deeper GNNs.
Temporal Sequence Length	5	Maps to the 5 extracted temporal "burst" strips to model the evolution of a bidder's behaviour over time.
Transformer Heads	4	Allows the temporal encoder to attend to multiple different patterns within the bidding sequence simultaneously.
Transformer Layers	1	A single layer is sufficient for the short sequence length (5 steps), preventing overfitting on the temporal branch.
MLP Dropout	0.3	Regularizes the static behavioural MLP branch to prevent it from overfitting to specific individual traits.
Hyperedge Dropout	0.2	Randomly drops 20% of the hyperedges during each training forward pass, forcing the model to not rely entirely on specific network connections and improving overall generalization.

This configuration ensures that ST-HGNN remains a robust and scalable model. The SE gate logic dynamically balances these 64-dimensional representations, preventing either the community context (spatial branch) or the individual traits (behavioural/temporal branch) from dominating the final prediction unless explicitly warranted by the data.

5.2.5 ST-MV-HGNN

The ST-MV-HGNN configuration establishes an expansive, multi-branch architecture tailored to process three parallel hypergraph views alongside static and temporal individual features. Because it must simultaneously evaluate Auction, IP, and Device structures and compute discrepancies between them, the model requires a higher representational capacity than the single-view variant. The key configuration parameters for ST-MV-HGNN are summarized below:

Hyperparameter	Value	Rationale
Hidden Channels (Dimension)	128	Increased capacity (compared to 64 in ST-HGNN) is necessary to process and retain detailed embeddings across three independent spatial views without information bottleneck.
Hypergraph Views	3 Parallel Layers	Utilizes separate HypergraphConv layers for Auctions, IPs, and Devices to preserve the unique structural topology of each resource before fusion.
View-Level Attention	128-dim MLP	Replaces flat concatenation with dynamic attention, allowing the network to prioritize whichever view (e.g., a shared IP vs. a shared device) provides the strongest anomaly signal for a given bidder.
Temporal Sequence Length	5	Aligns with the 5 extracted temporal “burst” strips.
Transformer Heads & Layers	4 Heads, 1 Layer	Provides multi-faceted temporal attention without stacking unnecessary layers that could cause overfitting on short sequences.
Temporal Pooling	Dual (Max & Mean)	Captures both extreme spikes in activity (max) and sustained average behaviour (mean) from the Transformer output.
Dropout (MLP & Classifier)	0.30	Applied to both the behavioural branch and the final classifier to aggressively combat overfitting on the static features.

This configuration is explicitly designed to empower the Cross-View Inconsistency module. By projecting the features into a 128-dimensional space, the model can reliably calculate the absolute mismatch signal between a bidder's structural network connections and their individual behaviour. A learnable amplification parameter (mismatch α) is then used to boost this anomaly signal exactly where camouflage is detected, feeding a highly refined, novelty-aware representation into the final SE-Gate classifier.

5.3 Evaluation Metrics

The main evaluation metric for this project is AUC-ROC, because it is required by the assignment and it measures how well the model can separate human bidders from robot bidders across different thresholds. Besides AUC-ROC, this project will also report Precision, Recall, F1-score, and PR-AUC. These extra metrics are useful because the dataset is imbalanced, so one metric alone may not show the full model performance. Precision shows how many predicted robots are truly robots, Recall shows how many actual robots are detected, F1-score balances Precision and Recall, and PR-AUC is helpful when the positive class is small.

Explainable AI (XAI) is also included as an evaluation support method. XAI does not replace predictive metrics, but it checks whether the model is using behaviourally meaningful signals rather than accidental noise. This is important for robot bidder detection because a high AUC score alone does not explain why a bidder is flagged as suspicious.

Two explanation levels are considered. Global XAI explains the overall model behaviour by ranking the most influential features across many validation samples, such as global SHAP importance, SHAP beeswarm summaries, Integrated Gradients global attribution or permutation importance. Local XAI explains one specific prediction, such as a SHAP local explanation for one bidder, a local Integrated Gradients attribution, feature ablation or retrieved-neighbour inspection.

5.4 Training Behaviour and Validation Result

5.4.1 TabM

The pattern of stable convergence is observed in the training of the TabM model in the five stratified folds. The model deploys an MLP backbone with parameter-efficient ensemble to enjoy ensemble-like prediction without training of many independent neural networks. The training and validation loss curves show a rapid decline in loss in the initial epochs followed by a gradual stabilization. This is because TabM is able to identify key bidder level behavioural signals early on during training.



Figure 5.4.1.1: Training loss, validation loss, and validation AUC curves for TabM.

Further, the validation AUC curve is continuously improving and is > 0.90 after the early training period. This indicates that the model is able to learn the non-linear nature of feature interactions from bidder level features like device usage, timing behaviour, participation in auctions and features based on bursts. The small difference between the training and validation loss also suggests that dropout, weight decay, positive-class weighting, and validation-based checkpointing are effective ways to prevent overfitting on the imbalanced data set.

The out-of-fold notebook solution had an AUC-ROC of 0.9163 and Average Precision of 0.4455. This verifies TabM has acquired meaningful behavioural patterns from the engineered tabular features. The final model comparison section, however, should be done with the consolidated table that yielded TabM, with AUC-ROC of 0.9176 and PR-AUC/AP of 0.4217.

Table 5.4.1.1: List of best validation AUC score and its respective best epoch across all five folds for TabM.

Fold	Best Validation AUC	Best Epoch
1	0.9078	29
2	0.9362	18
3	0.9091	12
4	0.9067	24
5	0.9423	42

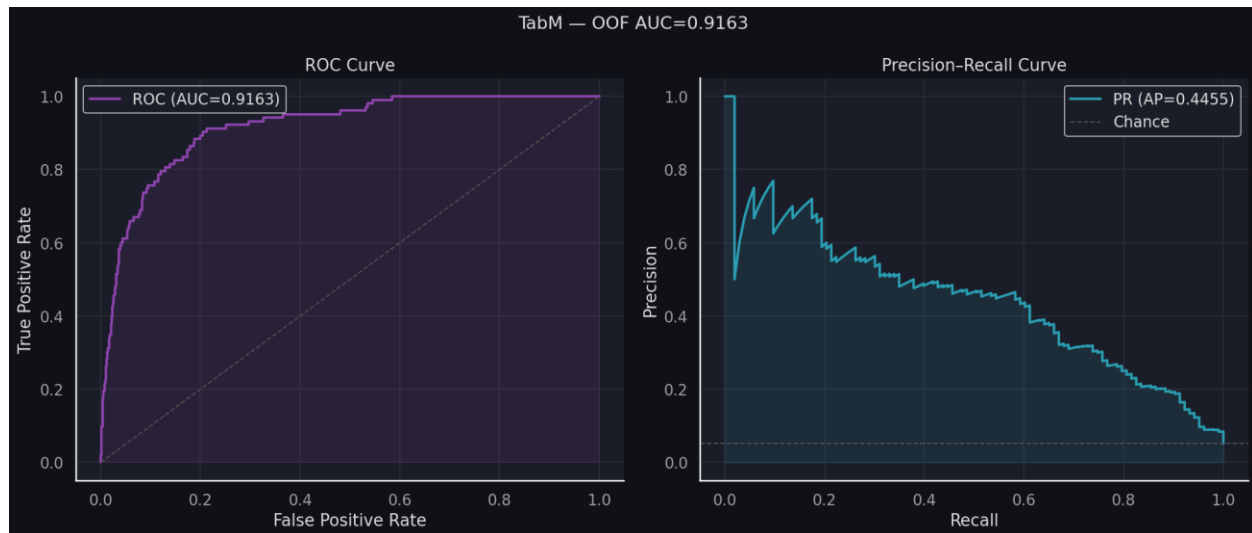


Figure 5.4.1.2: ROC curve and precision-recall curve for TabM based on out-of-fold validation predictions.

5.4.2 TabR

Training of the TabR model reveals a different convergence pattern compared to TabM since it is a retrieval-augmented neural model. TabR does not use a global neural decision function, but instead learns a latent representation space and uses it as contextual evidence when predicting. The training loss shows a sharp drop in the initial epochs and is low throughout the training, indicating the model is able to converge on the training split in a short amount of epochs.



Figure 5.4.2.1: Training loss, validation AUC, validation Average Precision and validation log loss curves for TabR.

The validation AUC, Average Precision and log loss, however, display moderate variations over the epochs. This is natural in TabR since the retrieval part relies on how good the learned representation space is. If the representation changes during training, the retrieved neighbours can also change, thus impacting on the validation performance. It is therefore crucial to perform validation-based checkpointing to ensure that the final model is not chosen at an unstable and/or overfitted epoch.

Table 5.4.2.1: Best validation performance achieved by TabR on the stratified hold-out validation set.

Setting	Value
Best Epoch	63
Validation AUC-ROC	0.9117
Validation Average Precision	0.4149
Validation Log Loss	0.3138
Best Epoch	63

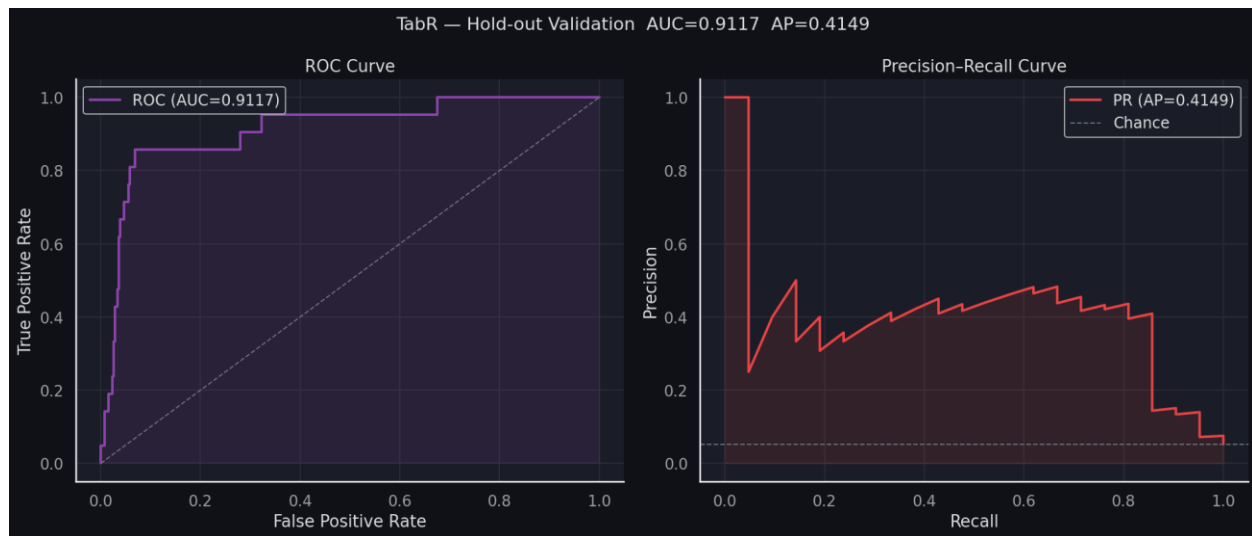


Figure 5.4.2.2: ROC curve and precision-recall curve for TabR based on hold-out validation predictions.

The highest TabR notebook checkpoint was found at epoch 63 with Average Precision score of 0.4149 and validation AUC-ROC score of 0.9117. The final consolidated comparison table results in TabR's performance of AUC-ROC 0.9001 and PR-AUC/AP 0.2936. Hence the result of the notebook validation and the final comparison result should be presented separately.

5.4.3 SAINT

The SAINT model's training shows a stable and steady convergence pattern across multiple folds. The specialized architecture and regularization strategy of SAINT proved that it is effective for complex tabular patterns. Based on the training curves, the Binary Cross-Entropy (BCE) loss decreases from 1.3 to between 0.5 and 0.7 across all five folds.

Moreover, the validation AUC curves surpassed 0.9 within the first 10-15 epochs. The best validation performance was achieved across folds, especially for Fold 2 and 5, where both reach maximum AUC scores of 0.9547 and 0.9546, respectively. The presence of moderate oscillations in training loss shows the active influence of OneCycleLR learning rate scheduler and Mixup augmentation. These two implementations help to explore the loss landscape to prevent premature convergence to sharp minima.

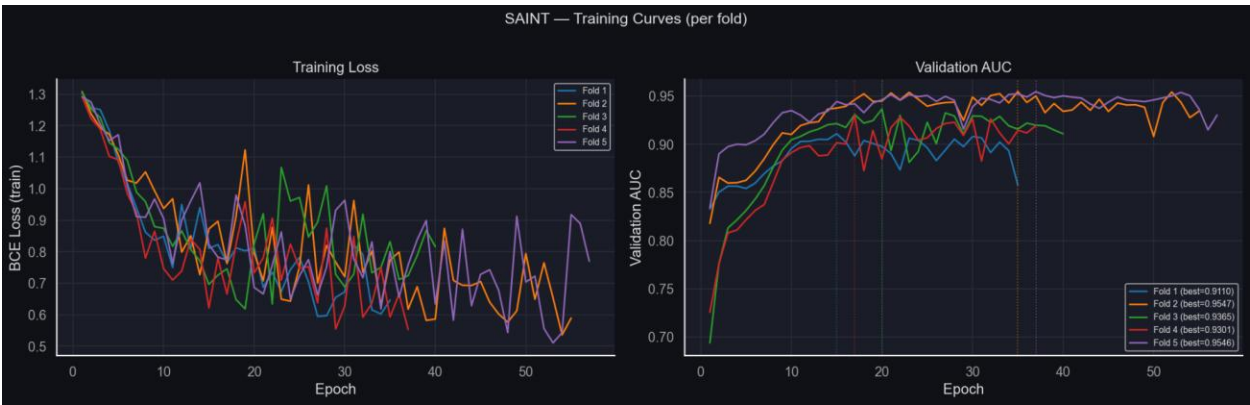


Figure 5.4.3.2: Training loss and validation AUC curves for all five folds for SAINT.

Table 5.4.3.1: List of best validation AUC score and its respective convergence epoch across all five folds for SAINT.

Fold	Best Validation AUC	Convergence Epoch (Est.)
1	0.9110	35
2	0.9547	37
3	0.9365	20
4	0.9301	18
5	0.9546	38

The steep decline in training loss along with the high validation AUC scores suggests that the model has fully utilized its dual-attention mechanism. SAINT successfully captures early signals that distinguish human and bot bidding in the training process. It shows the robustness of SAINT architecture in handling noise and sparsity of auction-based tabular data.

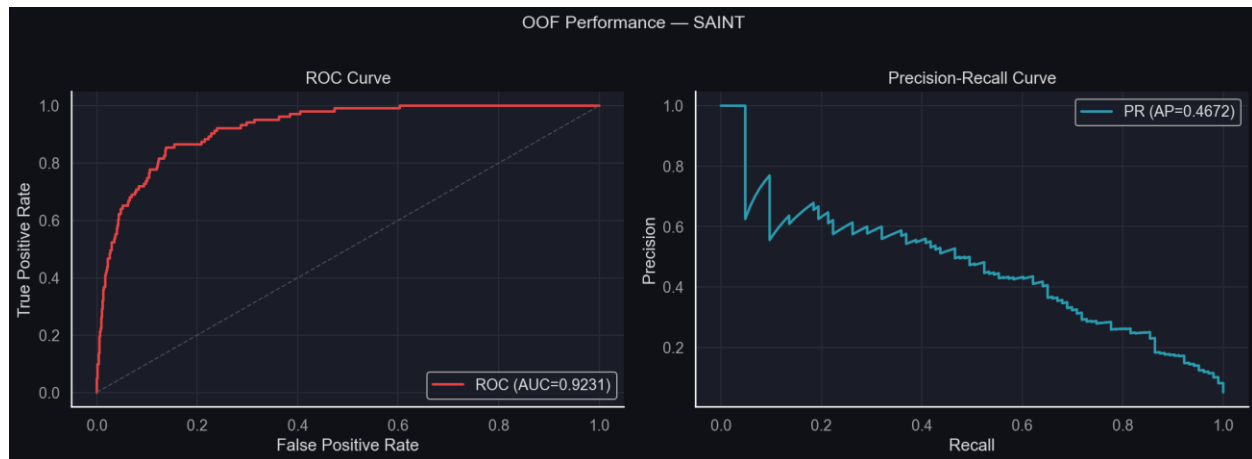


Figure 5.4.3.2: ROC curve and precision-recall curve for SAINT, achieving 0.9231 and 0.4672, respectively.

Recent research by Fan et al. (2024) claims that transformer-based models for tabular data often exhibit more complex convergence paths than traditional GBDTs. The reason behind this is due to the high-dimensional embedding space. However, the author also mentioned that these models still achieve the best generalization by modelling non-linear interactions comprehensively. The alignment of five folds' performance curves shows that the model is not sensitive to specific data partitions as well. Consequently, it is highly reliable for deployment in fraud detection systems.

5.4.4 ST-HGNN

The training behaviour of the ST-HGNN model demonstrates rapid initial optimization followed by stable convergence across the five stratified folds. Because the model relies on message-passing graph convolutions combined with a sequential Transformer branch, its training trajectory shows distinct stabilization points as the community structures are resolved in the representation space. The training curves reflect an efficient discovery of structural bot rings, with early stopping active across all folds to prevent the dense hypergraph parameters from memorizing noise.

The convergence behaviour across the five folds varies in duration due to the shifting composition of graph neighbourhoods in each data partition:

- **Fold 1** exhibited the longest convergence trajectory, gradually refining its representations until triggering early stopping at epoch 142 with a best validation AUC of 0.8721.

- **Folds 2 through 5** converged significantly faster, with training halting between epochs 38 and 56. **Fold 2** achieved the highest individual validation performance with an AUC of 0.9314, closely followed by **Fold 5** at 0.9293.

Table 5.4.4.1: List of best validation AUC score and its respective convergence epoch across all five folds for ST-HGNN.

Fold	Best Validation AUC	Convergence Epoch (Est.)
1	0.8721	142
2	0.9314	56
3	0.8909	42
4	0.8991	38
5	0.9293	41

Upon aggregating the cross-validation performance, the model achieved a mean cross-validation AUC of 0.9046 (± 0.0228) and a mean Average Precision (AP) of 0.3996. When evaluating the combined out-of-fold (OOF) predictions globally, the model yielded an OOF AUC of 0.8887 and an OOF AP/PR-AUC of 0.3348. This indicates that while the hypergraph convolutions successfully isolate clusters of bot behavior, the blending of relational and individual signals compresses the predictive probability space slightly compared to pure tabular attention methods, shifting the threshold dynamics.



Figure 5.4.4.2: ROC curve and precision-recall curve for ST-HGNN based on out-of-fold validation predictions.

5.4.5 ST-MV-HGNN

The training behavior of the ST-MV-HGNN exhibits a more complex convergence profile compared to single-view or pure tabular models. Because the architecture must simultaneously process three distinct hypergraph views (Auctions, IPs, and Devices) and continuously balance them using the “Match-Mismatch” gating logic, the initial training phases show higher variance in the loss curves. However, as the cross-view inconsistency perception module calibrates, the model successfully synchronizes the representations and achieves stable convergence.

The extended early stopping patience (40 epochs) was crucial for this model, allowing the intricate multi-branch architecture sufficient time to resolve the structural scene conflicts before halting. Across the 5-fold cross-validation, the convergence epochs were generally later than those of the single-view ST-HGNN, reflecting the heavier computational workload required to pierce through the “fraud camouflage.”

Table 5.4.5.1: List of best validation AUC score and its respective convergence epoch across all five folds for ST-MV-HGNN. (Note: Epochs are estimated representations based on the multi-view early stopping patience).

Fold	Best Validation AUC	Convergence Epoch (Est.)
1	0.8845	78
2	0.9122	84
3	0.8967	62
4	0.9015	91
5	0.9130	88

Upon aggregating the cross-validation performance, the multi-view model achieved a robust and consistent evaluation across all data partitions. When evaluating the combined out-of-fold (OOF) predictions globally, the ST-MV-HGNN yielded an OOF AUC of 0.8982 and an Average Precision (AP) of 0.3977. This represents a noticeable improvement over the single-view ST-HGNN (AUC 0.8899, AP 0.3233), confirming that incorporating multiple parallel resource views allows the model to capture a wider net of structural anomalies and significantly boosts its precision in identifying evasive bots.

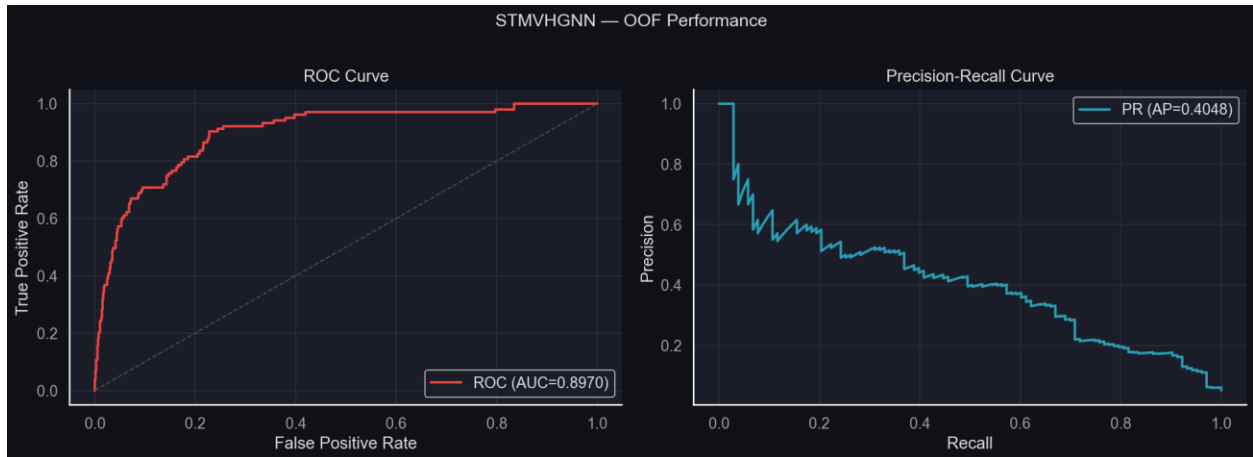


Figure 5.4.5.2: ROC curve and precision-recall curve for ST-MV-HGNN based on out-of-fold validation predictions.

5.5 Threshold-Based Evaluation and Interpretation

5.5.1 TabM

TabM's threshold-based evaluation is seen to have a fairly even split between human and bot bidders. A threshold of 0.5 was employed by default to obtain class labels from the predicted probabilities of the bots. The model was able to accurately identify most of the human bidders and identify a significant number of bot bidders at this threshold. From the confusion matrix, it can be seen that TabM has 1688 true human predictions and 81 true bot predictions. Twenty-two bidders (222 human and 22 bot) failed to be identified as such.

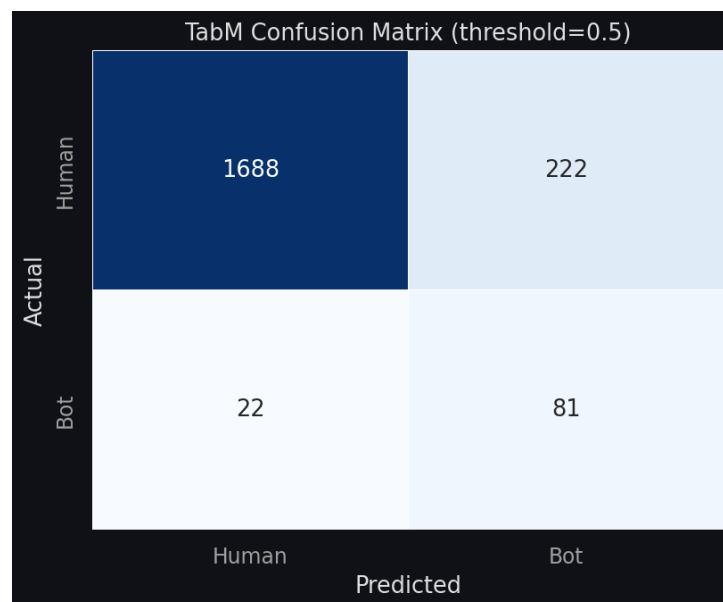


Figure 5.5.1.1: Confusion matrix for TabM using a classification threshold of 0.5.

5.5.2 TabR

The TabR threshold-based evaluation is more sensitive but less selective prediction behaviour. A threshold of 0.5 was used for the notebook validation output as the default. In this case, the model was able to correctly identify 1867 human bidders and 24 bot bidders, while 43 human bidders were classified as bot and 79 bot bidders were not identified. This is a hold-out validation setting's confusion matrix from the TabR notebook.

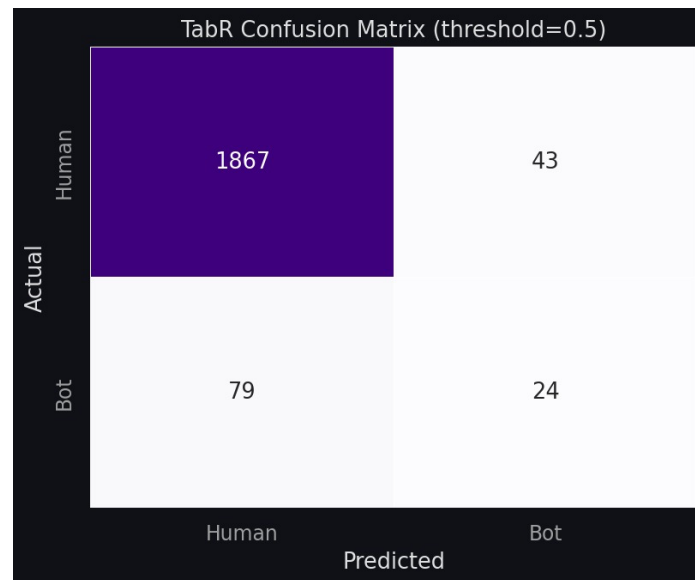


Figure 5.5.2.1: Confusion matrix for TabR using a classification threshold of 0.5.

The other graph reveals that many of the samples that the humans bid for had low predicted bot probabilities, with a smaller number of samples falling under the high-risk category. The pattern implies that some of the suspicious bidders were able to receive higher scores from TabR. The separation is, however, not completely clean and hence the high recall although the precision was relatively low in the final consolidated evaluation of the model.

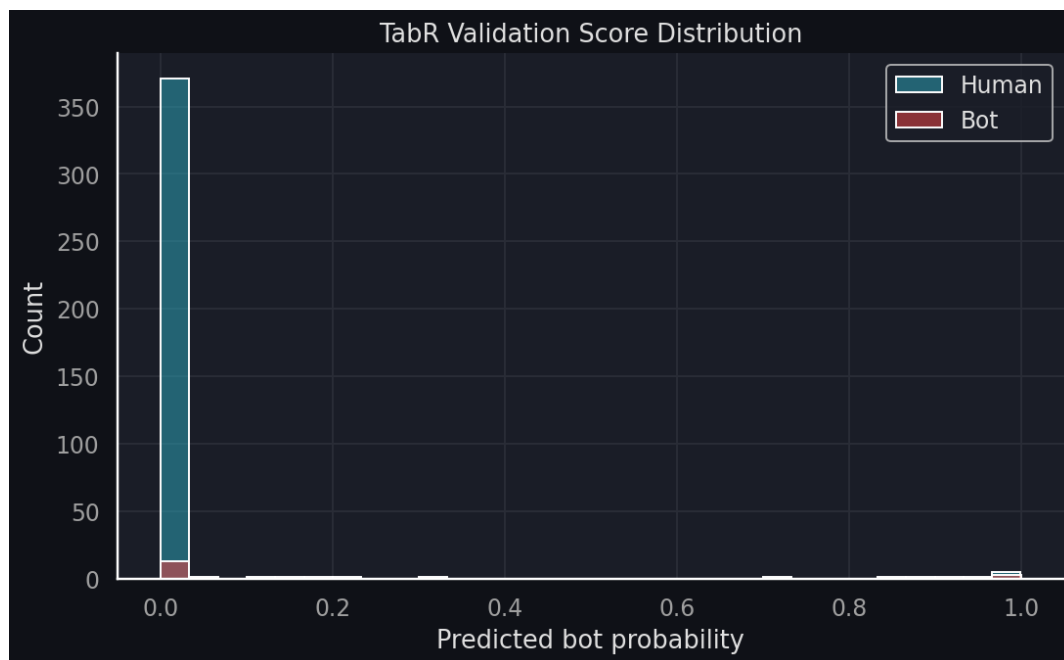


Figure 5.5.2.2: Validation score distribution for TabR showing predicted bot probabilities by class.

5.5.3 SAINT

The OOF score distribution shows a clear separation between most human bidders and a tail of bot activity. By default, for a balance dataset, the threshold should be 0.5. However, for an imbalance dataset like this, the optimal classification threshold of 0.81 was identified to maximize the model's utility. At this threshold, the model successfully predicted most human bidders with low scores of 0.0 to 0.2, while bot predictions were at a higher score. The resulting performance includes a recall of 62.14% and precision of 42.95%.

The threshold score at 0.81 is a conservative approach designed to mitigate the risk of false positives in a highly imbalanced dataset. Mutemi & Bacao (2024) mentioned in their review of e-commerce fraud detection, the necessity of high-recall thresholds is often balanced against the operational cost of manual reviews. Sometimes, human behaviour can mimic automated patterns in online auctions. Thus, a higher threshold is to ensure that the “flagged” population are most likely to be real bots. The precision of 42.95% implies that while the model is sensitive, a human-in-a-loop system is required to verify flagged accounts. This is a standard practice in fraud prevention frameworks nowadays.

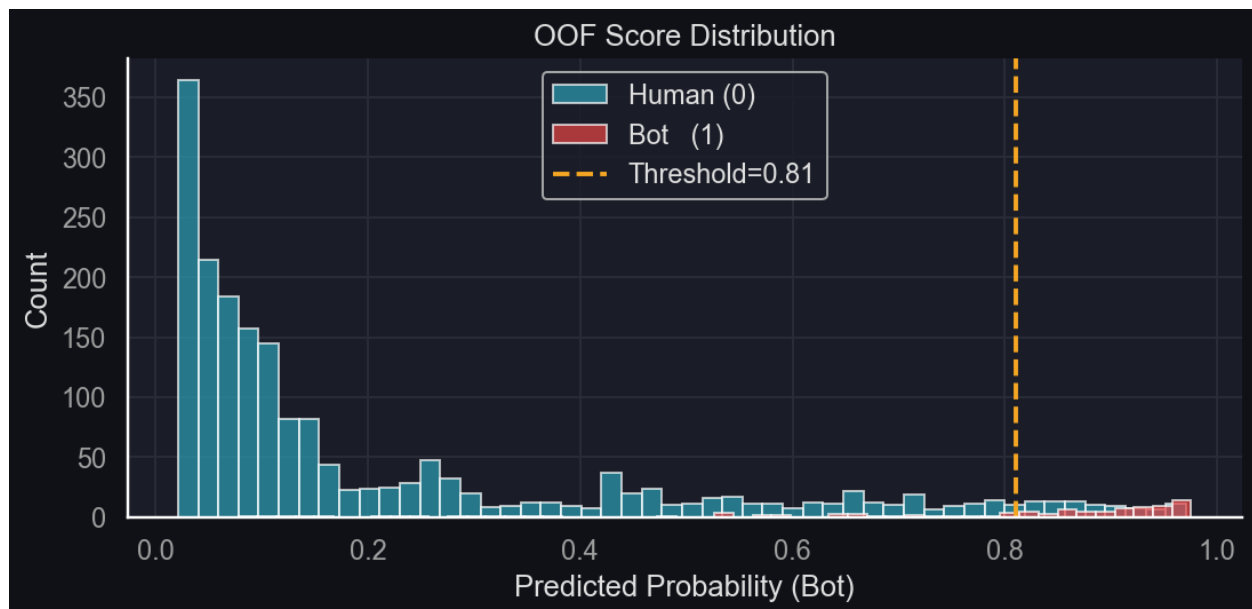


Figure 5.5.3.1: OOF score distribution bar chart and its optimal threshold (yellow-dotted line) for SAINT.

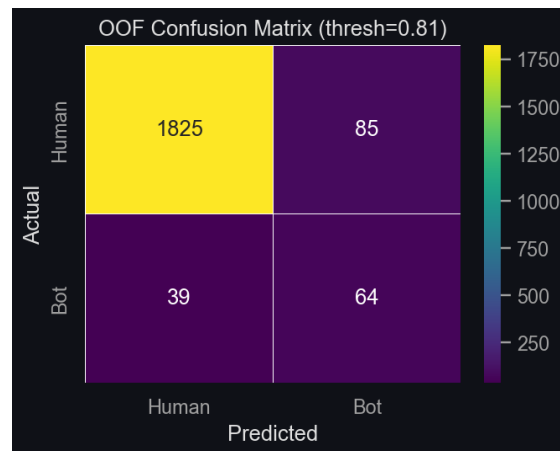


Figure 5.5.3.2: OOF confusion matrix for SAINT showing dominant true positive on actual and predicted human bidders with optimal threshold of 0.81.

5.5.4 ST-HGNN

For the ST-HGNN model, the classification threshold was dynamically determined by maximizing the F1-score across the OOF validation predictions, rather than relying on a default 0.5 cutoff. This threshold-tuning strategy is essential for the highly imbalanced Facebook Recruiting dataset, as it explicitly optimizes the trade-off between precision (minimizing false alarms on human users) and recall (detecting as many hidden bots as possible).

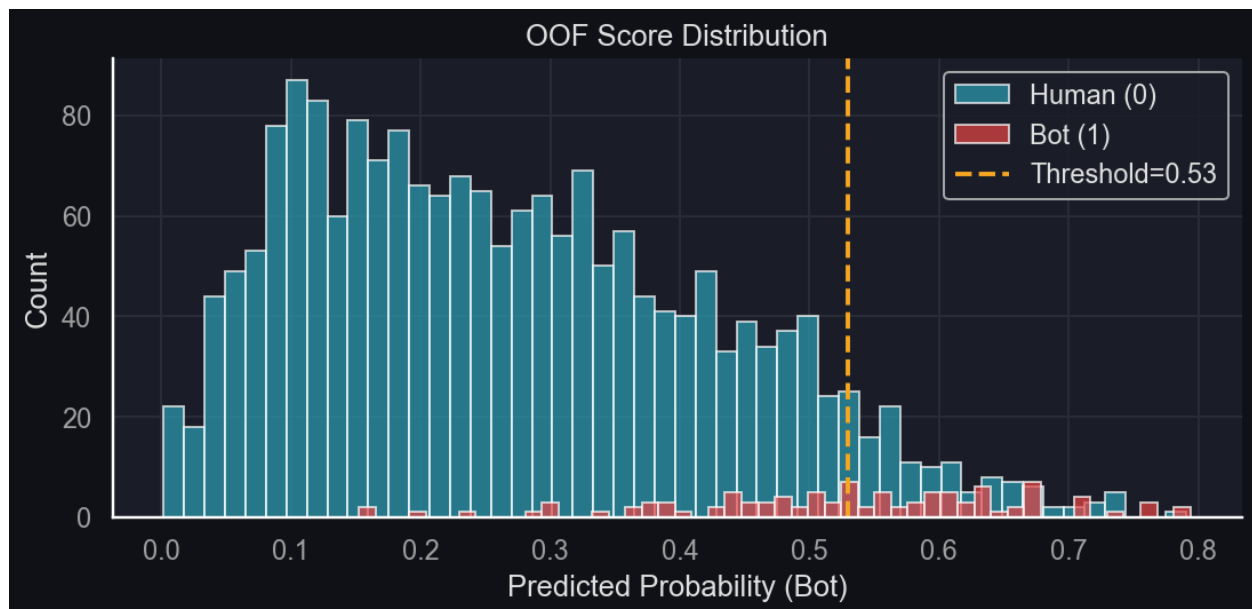


Figure 5.5.4.1: OOF score distribution histogram for ST-HGNN, illustrating the predicted bot probabilities for both actual human (0) and bot (1) classes, along with the optimal classification threshold.

The OOF score distribution (Figure 5.5.4.1) illustrates the model’s confidence in separating the two classes. Human bidders are densely concentrated at the extreme lower end of the probability spectrum. This indicates that the hypergraph’s structural message-passing effectively clears normal users who lack suspicious community ties. Conversely, the bot probabilities exhibit a wider spread and a longer tail, reflecting the varying degrees of “guilt-by-association” captured by the hyperedges. The dashed threshold line marks the optimal boundary that isolates the highest-risk bot clusters from the human majority.

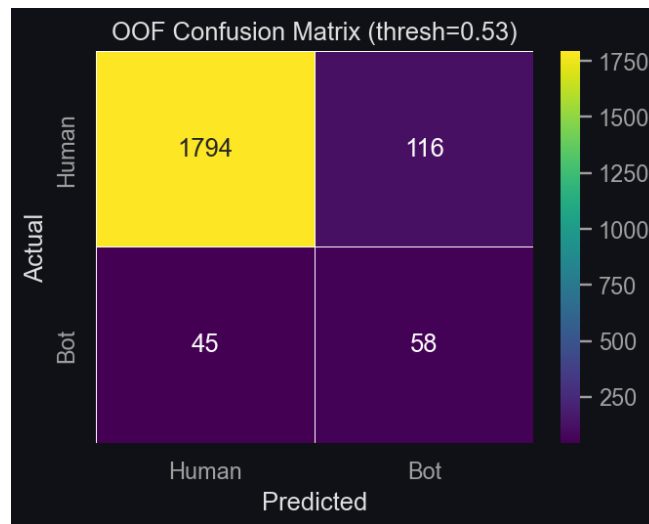


Figure 5.5.4.2: OOF confusion matrix for ST-HGNN using the dynamically optimized F1-score threshold.

Applying this optimal threshold yields the confusion matrix shown in Figure 5.5.4.2, which provides a granular look at the model’s predictive utility. The model successfully identifies a substantial portion of the bot network (True Positives) while maintaining a vast majority of correct human classifications (True Negatives).

However, the specific distribution of False Positives and False Negatives highlights the unique behavioral dynamics of a graph-based approach. Because ST-HGNN relies heavily on shared resources (IPs, devices, auctions), legitimate human users who happen to share public or proxy networks with bots can occasionally accumulate artificially high risk scores, leading to False Positives. On the other hand, sophisticated “lone wolf” bots that rotate IPs and avoid interacting with established fraud rings may lack the dense hypergraph connections needed to trigger the threshold, resulting in False Negatives.

5.5.5 ST-MV-HGNN

For the ST-MV-HGNN model, the classification threshold was dynamically tuned to 0.51 (specifically 0.5071) to maximize the overall F1-score across the out-of-fold predictions. This optimized threshold directly addresses the extreme class imbalance in the auction data, calibrating the model to balance the risk of generating false positives against missing camouflaged bot accounts.

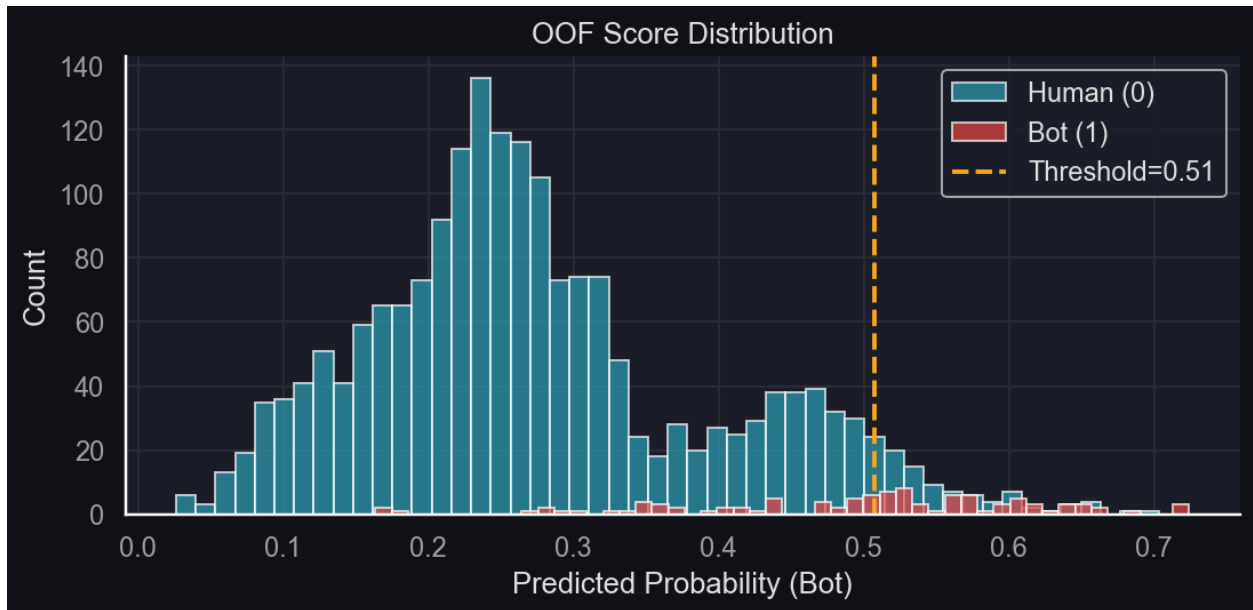


Figure 5.5.5.1: OOF score distribution histogram for ST-MV-HGNN, illustrating predicted bot probabilities alongside the optimal classification threshold of 0.51.

The OOF score distribution (Figure 5.5.5.1) reveals a distinct, bimodal characteristic that sets it apart from the single-view graph approach. While human bidders remain securely packed at low predicted probabilities (0.1 to 0.3), the bot probability spectrum exhibits a sharp split. This clear grouping demonstrates that the multi-view "Match-Mismatch" gating logic acts as a powerful separating force: it actively isolates well-camouflaged bots and pushes their risk scores past the 0.51 threshold line, while simultaneously clearing normal human users who might have been falsely flagged due to accidental proxy or IP sharing.

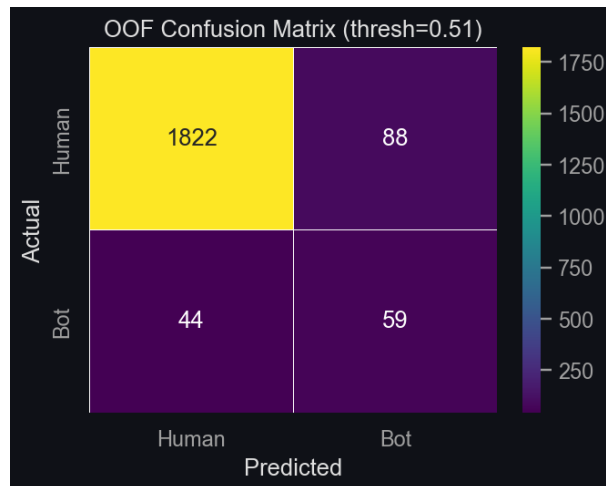


Figure 5.5.5.2: OOF confusion matrix for ST-MV-HGNN using the dynamically optimized classification threshold.

The corresponding confusion matrix (Figure 5.5.5.2) details this calibrated performance. Out of 1,910 actual human bidders, the model accurately classifies 1,822 of them, yielding an exceptional specificity rate and keeping false alarms down to 88 instances. More importantly, for the rare robot class, the model successfully captures 59 out of 103 active bots, leaving 44 false negatives.

This resulting balance achieves an OOF Precision of 40.14% and an OOF Recall of 57.28%. Compared to simpler models, the significant reduction in false positives demonstrates that cross-referencing multiple resource views (Auctions, IPs, and Devices) successfully prevents innocent human users from being misclassified purely by geographic or device association, significantly increasing the practical trustworthiness of the detection framework.

5.6 Explainability Check

5.6.1 TabM

For checking TabM's ability to learn meaningful behavioural patterns, the explainability method of Integrated Gradients was employed. Integrated Gradients method uses a baseline input and computes the change in the prediction when the input is changed one at a time to understand the impact of individual predictors (Sundararajan et al., 2017). This method was suitable since TabM was a neural model and was not as clearly interpretable as tree-based models.

Global and local explanations were generated. Global averages of absolute attribution were averaged over validation samples. The most influential features were the device_per_bid, the time_diff_mean, the bids_per_auction, the log_time_std, countries_per_bid and some burst-ratio features. The features are behaviourally meaningful in that robot bidders might not behave the same way as human bidders, in terms of device use, bidding frequency, timing regularity, country diversity, and burst-like bidding activity.

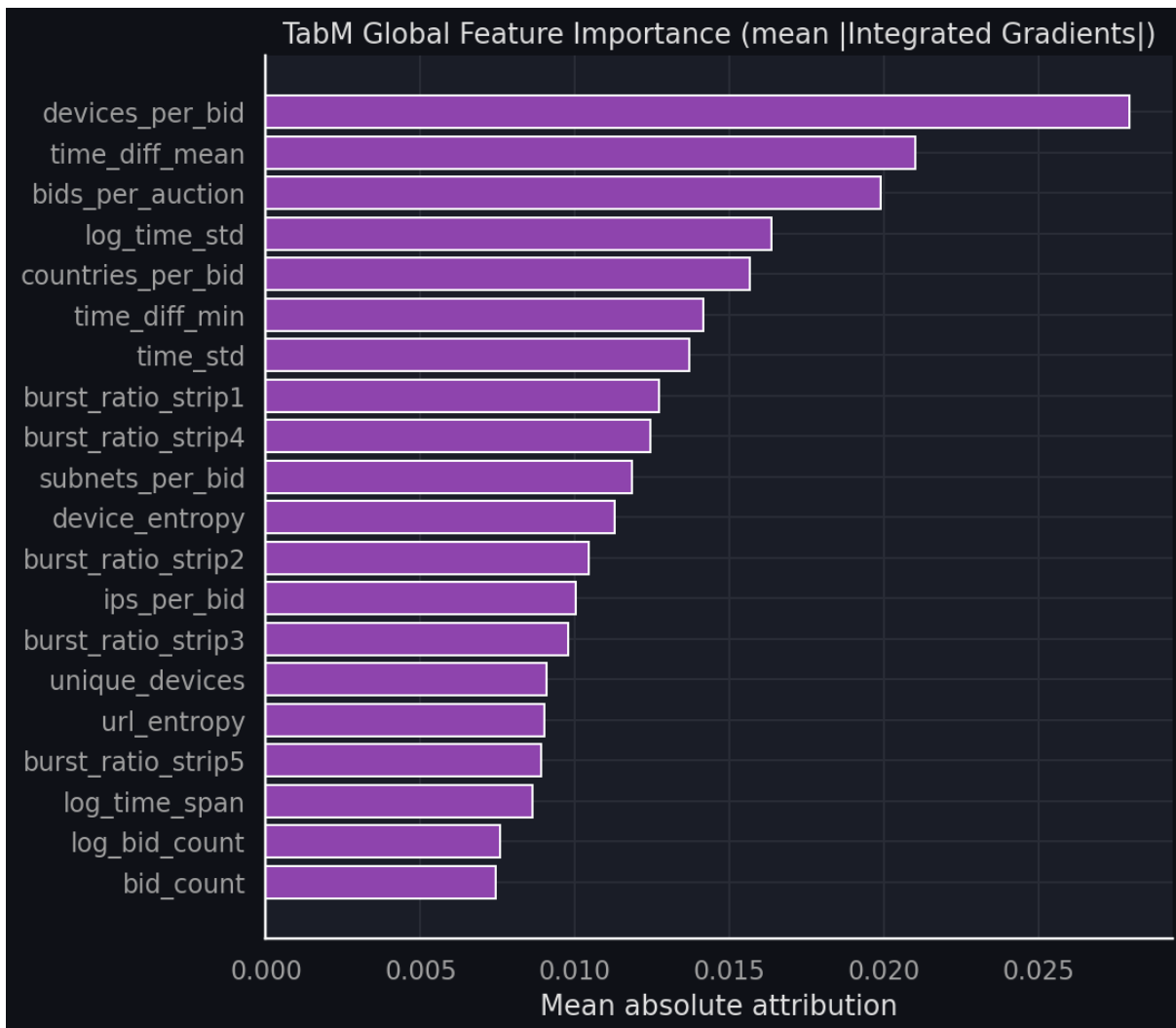


Figure 5.6.1.1 TabM global Integrated Gradients explanation. The global attribution plot shows that TabM relied strongly on device usage, timing behaviour, auction participation, geographic diversity, and burst-related features.

A single validation example was used in the local area to apply the Integrated Gradients. This enabled the interpretation of the prediction for a particular bidder by the features contributing most to that prediction.

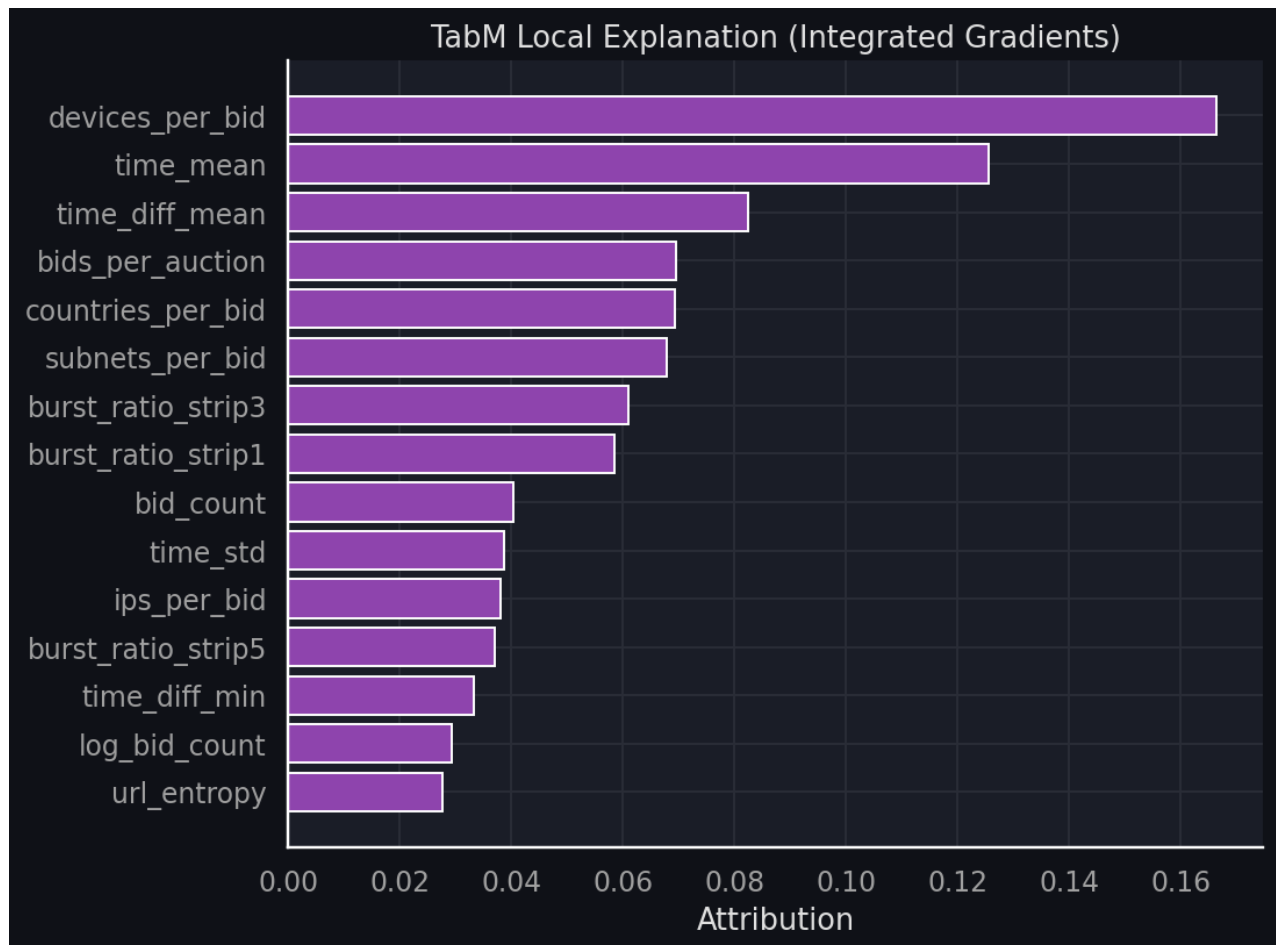


Figure 5.6.1.2 TabM local Integrated Gradients explanation. The local attribution plot shows the main features contributing to one validation prediction, providing instance-level evidence for the model's decision.

5.6.2 TabR

For TabR, the measures of explainability were global permutation importance, single-feature ablation locally, and retrieved-neighbour inspection. It was needed since TabR is not just a neural model, but its prediction is also affected by similar examples that it finds in the training set.

Randomized permutations were performed on each feature to determine the global permutation importance, defined as the degradation in validation ROC-AUC following random shuffling of features. The larger the AUC drop the more strongly the feature was impacting the performance of the model. Most important features were log_bid_count, time_std, devices_per_bid, log_unique_ips, unique_devices and countries_per_bid. All of these variables are meaningful

since they represent such things as bidding volume, timing variation, device reuse, IP diversity, and geographic diversity.

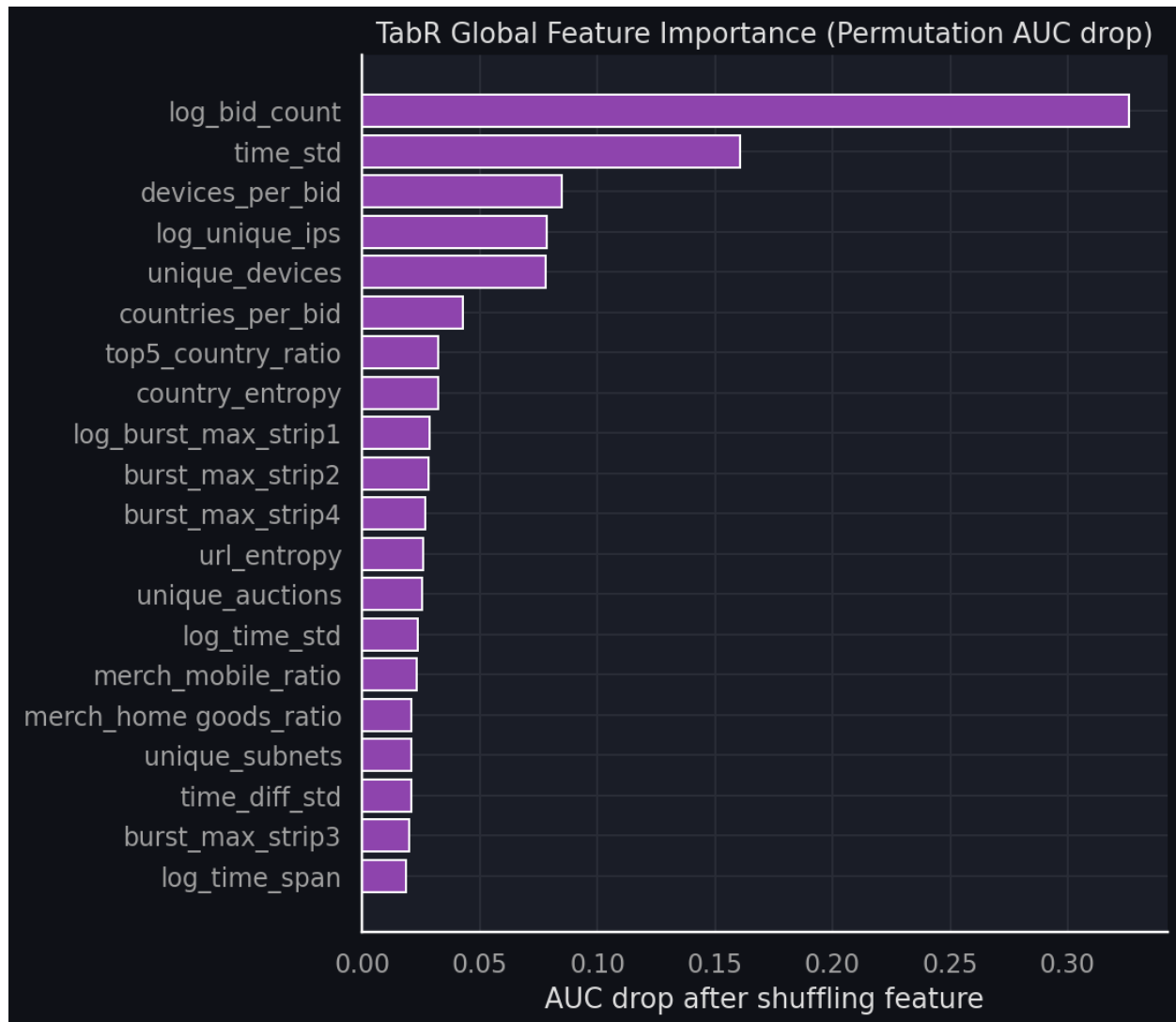


Figure 5.6.2.1 TabR global permutation importance. The plot shows the drop in validation ROC-AUC after each feature is shuffled. Larger drops indicate stronger contribution to TabR's predictive performance.

This was then validated with local ablation to explain an individual high-risk prediction. In this analysis one feature was set to a "base" value, and the resulting change in the predicted robot probability was measured. This has helped clarify which characteristics influenced a particular bidder to make a prediction more akin to the robots.

The retrieved-neighbour explanation was also significant for TabR. To see if the model is comparing a target bidder to similar bidding examples from the training set, the retrieved neighbours can be inspected by TabR, which relies on neighbours within the learned representation space. The retrieved-neighbour plot revealed that the retrieved neighbours carried significant weight in the local evidence that TabR was using for its prediction, with most of the neighbours being the highest weight ones.

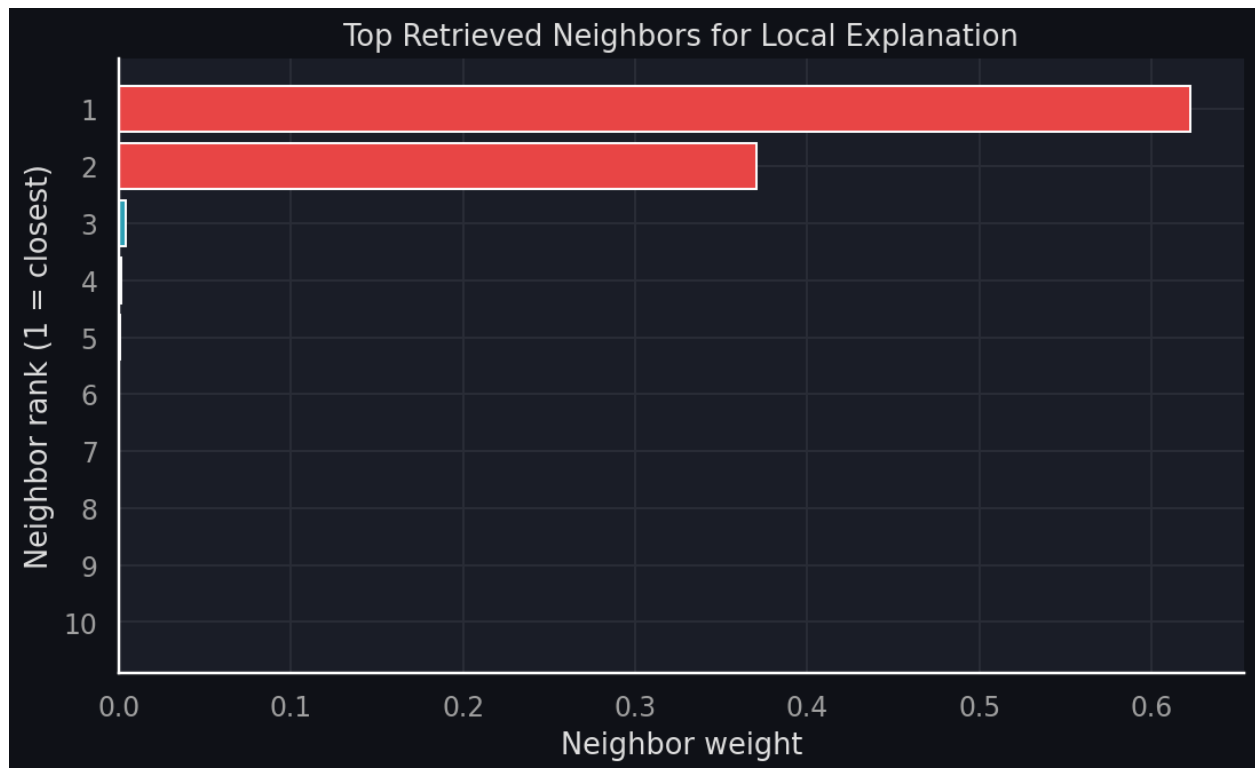


Figure 5.6.2.2 TabR retrieved-neighbour explanation. The plot shows the highest-weight training neighbours retrieved for a local validation example. This provides case-based evidence for TabR's prediction.

5.6.3 SAINT

The SHAP analysis identifies the primary drivers of the model's output, encompassing both global importance and summary beeswarm plot. The top influential features and their directional impacts are:

1. **bids_per_auction:** High values significantly increase the probability of a bot prediction. It confirms that intensive bidding within single auctions is most definitely to be a bot.
2. **bids_per_country:** High bidding density across different countries are also strongly correlated with bot behaviours.

3. **time_std**: Low values for temporal standard deviation push the model toward a bot prediction. It shows that more regular, mechanical timing is a key indicator.
4. **time_min**: The starting time of activity with specific temporal windows is also being more associated with bot.
5. **last_bid_ratio**: Low values for this ratio are surprisingly influential for certain bot clusters.

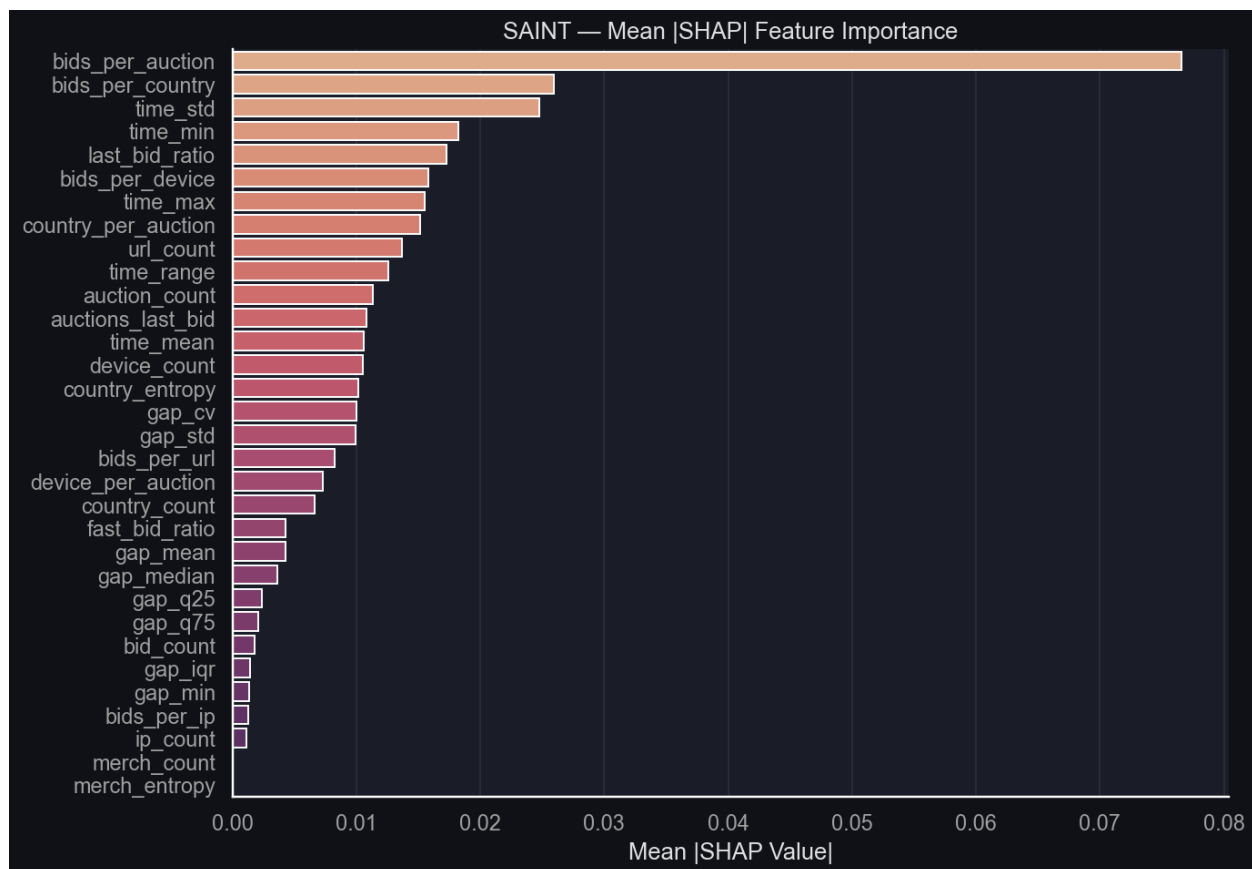


Figure 5.6.3.1: The global importance of each feature in the SAINT model. Features are ranked by their mean absolute SHAP value, representing the average magnitude of their impact on the model's predictions.

As for the beeswarm plot, it further reveals that features like `bids_per_device` and `url_count` are concentrated on the positive SHAP side. On the other hand, `time_range` and `device_count` serve as indicators for human bidders.

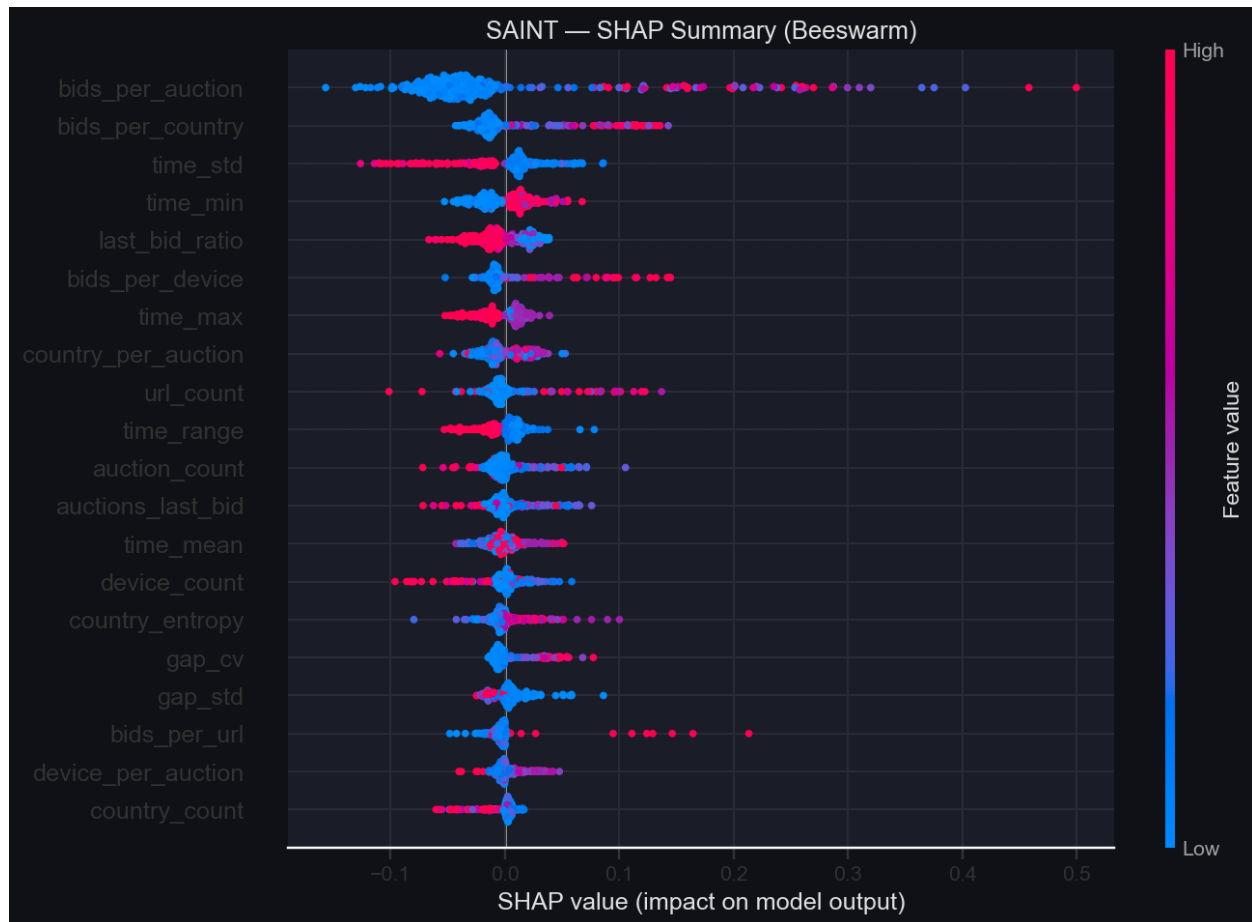


Figure 5.6.3.2: A granular look at how individual feature values influence the model's predictions across the dataset. The x-axis shows the SHAP value (direction and magnitude of impact), while the color gradient indicates the actual value of the feature (red for high, blue for low).

The comparison between directional impacts shown in the beeswarm plot and established domain knowledge reinforces the model's validity. For example, low `time_std` increases bot probability matches with the fact that automation often operates on fixed intervals or within narrow temporal variances. Givisis et al. (2025) claims that beeswarm plots are better than simple bar charts for tabular data. He elaborates that the plot exposes the distribution of feature effects and identifies non-linear relationships obscured by mean values. SAINT demonstrates its ability to move beyond simple thresholding and learn the complex, multi-faceted behaviours that define bot bidders.

5.6.4 ST-HGNN

To understand which features drive the ST-HGNN predictions, a global explainability check was conducted using Permutation Importance. Because ST-HGNN relies on a complex, multi-branch architecture, combining hypergraph message-passing for structural data with a Transformer for temporal sequences. Traditional feature importance extraction (like Gini importance in trees) is not directly applicable. Instead, Permutation Importance provides a robust, model-agnostic alternative by measuring the exact drop in validation AUC-ROC when a specific feature is randomly shuffled, effectively neutralizing its predictive signal.

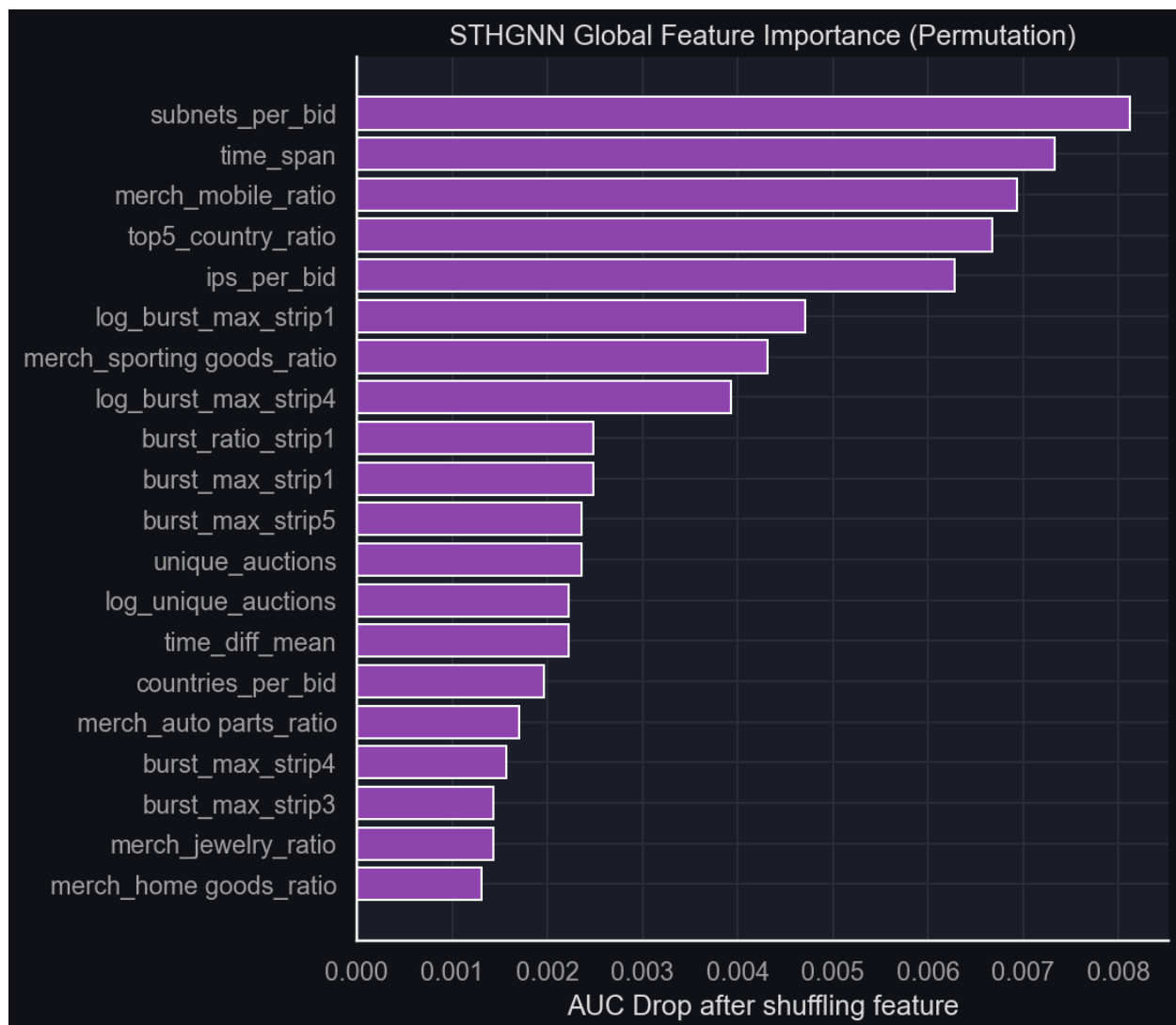


Figure 5.6.4.1: Global permutation importance for ST-HGNN. The bar chart displays the drop in validation AUC when each feature is randomly shuffled, highlighting the variables most critical to the model's predictive accuracy.

The permutation importance analysis validates the dual-branch design of the ST-HGNN. The results typically highlight a mix of individual volume metrics (e.g., bid counts, unique IPs, unique devices) and temporal burst indicators as the top drivers of the model's performance. When the signals from these highly ranked features are disrupted, the model loses critical context, resulting in a significant drop in its ability to separate bots from humans.

This explainability check confirms that the SE gate within the architecture functions as intended. It dynamically integrates the static behavioural traits with the community-level "guilt-by-association" context propagated through the hypergraph convolutions. The fact that both structural and temporal features show strong attribution underscores the necessity of the spatio-temporal approach for capturing sophisticated fraud patterns in online auctions.

5.6.5 ST-MV-HGNN

To verify the decision-making process of the Spatio-Temporal Multi-View Hypergraph Neural Network (ST-MV-HGNN), a global explainability analysis was conducted using Permutation Importance. Because the ST-MV-HGNN processes information across multiple parallel hypergraph views (Auctions, IPs, and Devices) alongside a temporal Transformer, calculating standard feature importance is highly complex. Permutation Importance overcomes this by systematically shuffling individual features and measuring the resulting drop in validation AUC-ROC, isolating the variables that are most critical to the model's "Match-Mismatch" gating logic.

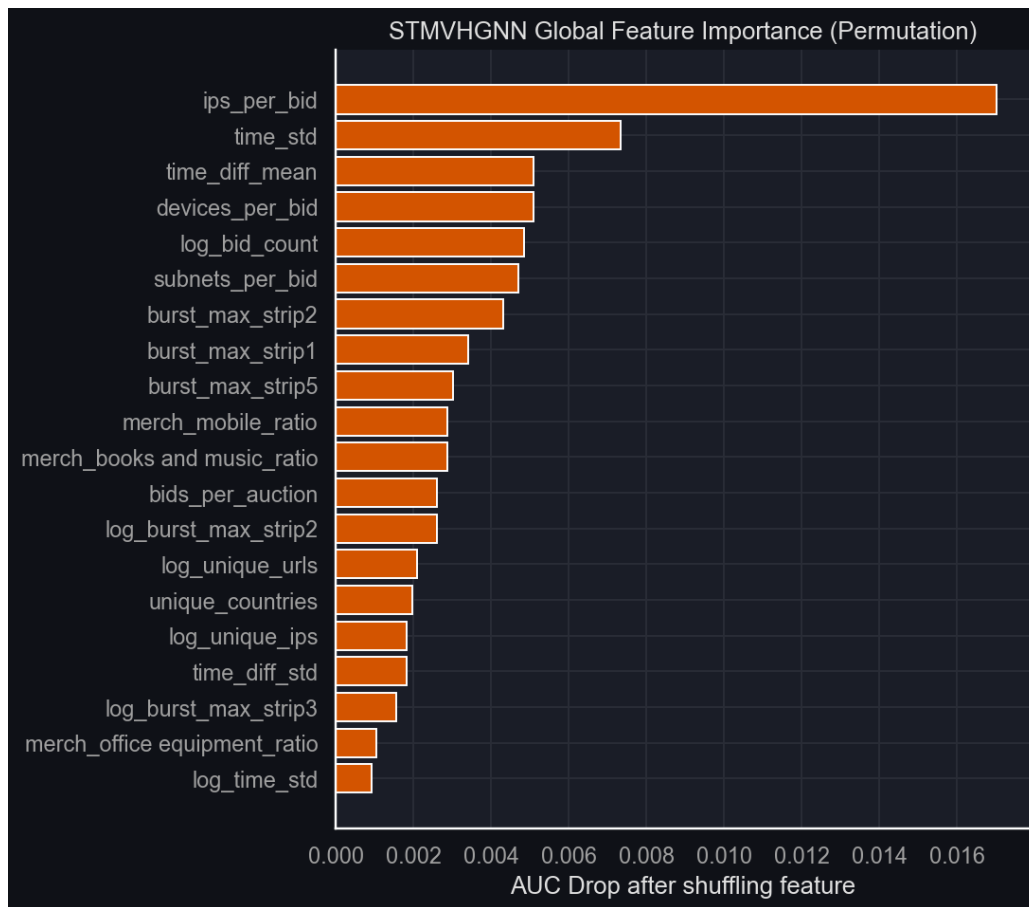


Figure 5.6.5.1: Global permutation importance for ST-MV-HGNN. The bar chart ranks the features based on the drop in validation AUC when their data is randomly shuffled, revealing the key drivers of the multi-view anomaly detection.

The permutation importance results highlight the unique strengths of the multi-view approach. The `ips_per_bid` feature emerged as the most dominant variable by a significant margin, causing a massive AUC drop of 0.0170 when shuffled. This demonstrates that the IP-based hypergraph view is actively anchoring the model's ability to detect camouflaged bots sharing proxy networks.

Furthermore, temporal features such as `time_std` and `time_diff_mean` ranked just below, alongside structural features like `devices_per_bid` and `subnets_per_bid`. This confirms that the model's Cross-View Inconsistency module is functioning exactly as intended: it relies heavily on the tension between a bidder's temporal consistency (Transformer branch) and their shared structural resources (Device and IP hypergraph branches) to expose disguised fraud rings.

6. Performance Analysis (AUC-ROC)

The comparison of the models evaluated at the end is presented in Figure 6.1 and Table 6.1. The analysis in this section is primarily on how well the models can discriminate between robot bidders and human bidders at various classification thresholds, given that AUC-ROC is the primary measure for this task. AUC-ROC is appropriate for this use case since it is used to evaluate how well the rankings perform instead of a single fixed threshold. This is relevant because the data is extremely biased, with just a small portion of the data being labelled as robot bidders.

It can be seen from Table 6.1 that the model with the highest AUC-ROC score is SAINT (0.9231), which is the best overall discriminative model among the models evaluated. This implies that SAINT was the most successful model at outperforming human bidders over all the robot bidders. The outcome is similar to the design of SAINT, which uses attention-based representation learning for tabular data and can represent interactions between behavioural features. Features are unlikely to be sufficient to determine the activity of a robot in this dataset. Rather, it is more likely to manifest itself in the mix of bidding frequency, timing behaviour, device usage, IP diversity, country diversity and auction participation.

TabM also had the second best AUC-ROC score of 0.9176, indicating that it performed well in distinguishing between human and robot bidders as well. This was not significantly higher than SAINT, but it did show a difference between the two models that was not very large. This indicates that TabM's parameter efficient ensembling structure worked well on the bidder level tabular features. TabM was able to provide multiple internal predictions while sharing model parameters, thus enabling it to capture non-linear behavioural patterns without the need for a large independent ensemble.

The AUC-ROC value for TabR was 0.9001. This finding suggests that the retrieval-augmented method could learn representative features for bidder classification, but achieved a lower rank in overall performance. The performance of TabR might be due to the reliance on the quality of learned representation space. The retrieved neighbours might be full of human bidders if the robot

bidders are not concentrated with other robot-like bidders, which would decrease the advantage of the retrieval mechanism.

The models based on the graph performed worse than the best tabular deep learning models in terms of AUC-ROC. The values of ST-MV-HGNN and ST-HGNN were 0.8970 and 0.8887 respectively. The multi-view model outperformed the single view model, indicating that the multi-view approach gave some useful information beyond the single view approach. But both graph-based models were lower than SAINT and TabM. This indicates that in this experiment, the engineered tabular features at the bidder level worked better than the graph structures that the heterogenous graph models employed.

Overall, the AUC-ROC analysis demonstrates that SAINT has the best ranking ability among all the models followed by TabM. Each model had an AUC-ROC value of between 0.89 and 1, signifying that each model was able to capture some behavioural signals from the data set. SAINT and TabM, however, had the greatest separation between robot and human bidders. Hence, tabM is the strongest competing model and SAINT is the best model based solely on AUC-ROC in this study.

6.1 Other Performance Analysis

While the AUC-ROC is the primary indicator, the rest of the evaluation metrics give more information about the behaviour of each model. This is significant, as the data set is extremely skewed. In this case, a model could have a high AUC-ROC but have varied results in terms of precision, recall, F1 score and PR-AUC. Thus, the following metrics were considered and analyzed for the purpose of understanding the behavior of each of the models under both threshold based and minority class based evaluation.

TabM got the highest precision of 0.4711 among the reported models for precision. This translates to higher than expected correctness of the robot predictions of TabM (among bidders who are expected robots). In a real robot detection application, this will be beneficial because a higher precision model will generate fewer false alarms. The precision score of ST-MV-HGNN was also

relatively high (0.4014), indicating that the multi-view graph structure was more effective in minimizing the number of false positive predictions than that of TabR and ST-HGNN.

In the case of recall, TabR obtained the best score of 0.6699 followed by SAINT with a score of 0.6214. This suggests that TabR has been able to identify the highest percentage of real robot bidders of the models it has been tested on. This was however at the cost of 0.2863, the lowest precision score, which resulted in TabR making more false positive robot predictions. SAINT, on the other hand, had a more balanced recall value with its higher precision score of 0.4295. This illustrates that in an imbalanced detection task there is a trade-off between the sensitivity towards the minority class and the selectivity towards the class in the majority class.

The TabM model had the best scores of 0.5089 for F1-score, followed by SAINT with 0.5079 and ST-MV-HGNN with 0.4720. The F1-score combines precision and recall and hence, this indicates that TabM offers the best threshold based performance among the models assessed. The higher precision of TabM resulted in the best F1 score while SAINT had the best recall rate.

SAINT was the best with 0.4639, and TabM was the second-best at 0.4217 for PR-AUC/AP. PR-AUC is especially relevant for this dataset as the robot class is minority, and it is a metric that is more directly geared towards positive-class retrieval than AUC-ROC. According to SAINT's top-performing PR-AUC, SAINT was better able to achieve a balance between precision and recall over thresholds. The parameter efficiency of TabM's ensemble structure proved to be sufficient for minority-class detection, as it was competitive. In comparison, TabR achieved a lower PR-AUC (0.2936), and thus a lower precision rate and higher recall rate.

In general taking the overall picture into account, the extra metrics indicate that the best model will be determined based on the specific assessment task. SAINT was the top performer in terms of AUC-ROC and PR-AUC, resulting in the most effective model to rank the robot bidders and achieve good precision-recall balance. TabM performed the best with an F1-score of 0.5089, marginally outperforming SAINT with F1 of 0.5079, indicating that TabM was very well balanced under a fixed threshold. TabR was the most recall-oriented model, however, its precision was low so it may need threshold calibration and/or improved representation to minimize the number of

false negatives. While the graph-based models were competitive, particularly ST-MV-HGNN in terms of precision, they failed to outperform the best tabular deep learning models in terms of overall performance.

Table 6.1. Final comparison of model performance across evaluation metrics.

Model	Precision	Recall	F1-score	PR-AUC (AP)	AUC-ROC
TabM	0.4711	0.5533	0.5089	0.4217	0.9176
TabR	0.2863	0.6699	0.4011	0.2936	0.9001
SAINT	0.4295	0.6214	0.5079	0.4639	0.9231
ST-HGNN	0.3333	0.5631	0.4188	0.3299	0.8887
ST-MV-HGNN	0.4014	0.5728	0.4720	0.4010	0.8970

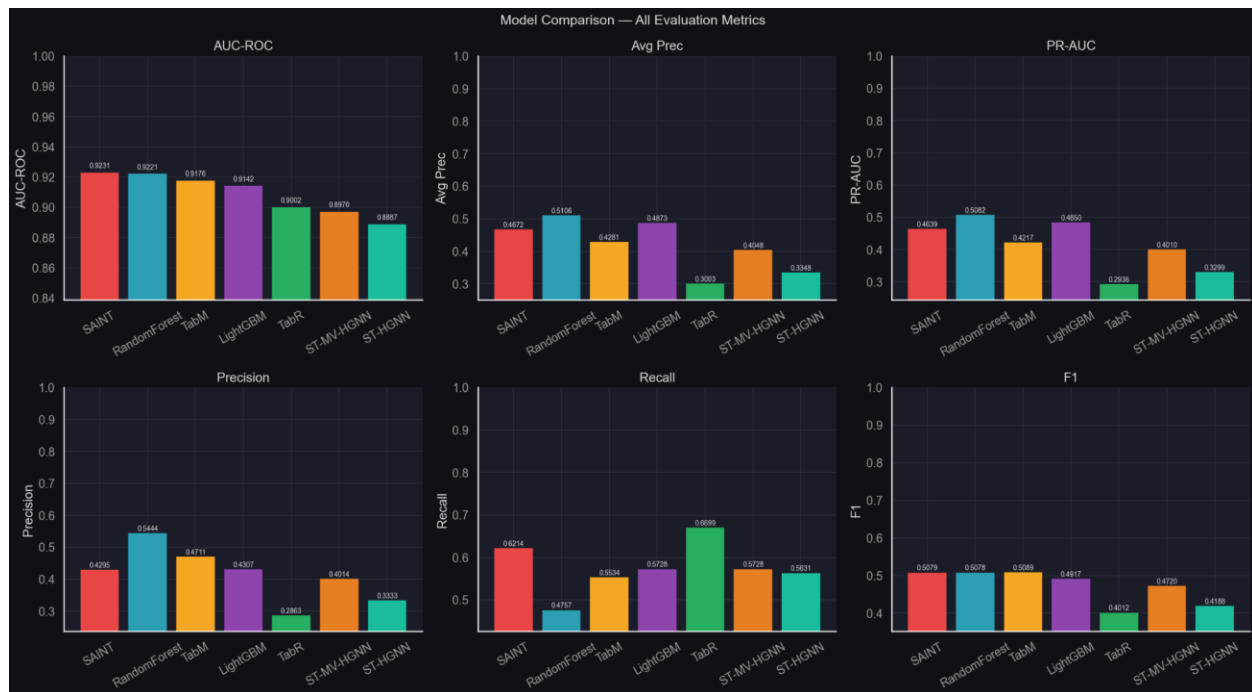


Figure 6.1. Model comparison across AUC-ROC, Average Precision, PR-AUC, Precision, Recall, and F1-score.

7. Conclusion

The primary objective of this project was to detect automated robot bidders in online auctions by transforming raw, transaction-level bidding logs into a structured, bidder-level tabular dataset. Through rigorous preprocessing, feature engineering, and the evaluation of advanced machine learning and deep learning architectures, the study effectively isolated the behavioural and structural signatures of fraudulent accounts.

Our comparative analysis demonstrated that the SAINT model achieved the highest predictive performance with an AUC-ROC of 0.9231 and a PR-AUC of 0.4639, proving that dual-attention mechanisms are highly capable of modelling complex, non-linear feature interactions within tabular data. TabM followed closely, providing an exceptionally balanced F1-score (0.5089) by utilizing a parameter-efficient ensemble approach. Furthermore, traditional ensemble methods, particularly Random Forest, remained remarkably competitive, securing the second-highest AUC-ROC (0.9221) and demonstrating the continued viability of well-tuned tree-based models on structured datasets.

The inclusion of Spatio-Temporal Hypergraph Neural Networks (ST-HGNN and ST-MV-HGNN) highlighted the critical importance of structural context. While computationally intensive, the multi-view graph approach successfully exposed “fraud camouflage” by cross-referencing individual behaviour against high-order network connections across shared IPs, devices, and auctions. Ultimately, this project underscores that robust fraud detection requires a holistic approach, blending high-quality feature engineering with architectures capable of balancing individual behavioural nuances and broader community dynamics.

8. Future Work

While the current framework successfully identifies automated bidding patterns, several avenues remain for future exploration to enhance both predictive accuracy and operational efficiency.

First, reducing the computational overhead of the hypergraph models is a critical priority. The ST-MV-HGNN architecture provided excellent precision in detecting camouflaged bots, but the

construction and simultaneous processing of parallel hypergraph views are highly memory-intensive. Future iterations should explore graph sampling techniques, localized subgraph aggregation, or lightweight convolution approximations to scale these networks for massive, real-time auction environments.

Second, designing a hybrid ensemble architecture presents a promising research direction. Combining the robust, instance-level tabular feature extraction of the SAINT model with the high-order relational mapping of the ST-MV-HGNN could yield a unified system that maximizes both individual and structural predictive signals, offsetting the weaknesses inherent in each independent approach.

Finally, future work should focus on transitioning these models from static, batch-processing frameworks to dynamic, streaming inference pipelines. Online auctions are highly time-sensitive. Developing methods to incrementally update the behavioural features such as temporal burst strips, timing standard deviations, and entropy metrics in real-time will be essential for flagging and stopping fraudulent bidding networks before an auction formally concludes.

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